Hyperspectral Image Classification via Dynamic Adaptive Graph Convolutional Network

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***Abstract*—A dynamic graph convolutional network (DGCN) can represent temporal evolutionary features. Its compatibility with the spectral-dimensional characteristics of hyperspectral images (HSIs), such as continuous gradients and local abrupt changes in spectral signatures, makes it valuable in this field. This study introduces dynamic graph modeling into HSI analysis. By mapping spectral dimensions onto virtual time series, we propose the dynamic adaptive graph convolutional network (DAGCN). The core idea is to use dynamic graph evolution to simulate continuous variations and local abrupt changes in spectral sequences, thereby capturing subtle spectral features of the Earth's surface. The framework includes three components: the dynamic graph sequence (DGS) construction strategy, which uses superpixel segmentation and spatiotemporal graph modeling to construct slice graphs for each spectral band. The multiscale adaptive graph convolution (MAGC) module dynamically generates multiscale adjacency matrices through adaptive convolutions, learning pixel features within homogeneous regions while preserving multiscale context. The spatio-temporal graph collaborative fusion (STGC) strategy includes a dynamic segmentation-based graph update (DSG Update) module and a multispectral channel attention (MSCA) mechanism. DSG Update dynamically optimizes the graph structure of adjacent temporal phases using MAGC's adjacency matrix, extending independent multigraph learning to spatiotemporal coupled multigraph learning for modeling spectral–spatial evolution. MSCA assigns self-attention weights to each temporal phase to enhance discriminative band selection. Experiments on public HSI datasets show that DAGCN outperforms state-of-the-art methods in classification accuracy, especially in capturing subtle spectral variations, confirming its effectiveness and superiority.**

***Index Terms*—Attention mechanism, dynamic graph convolutional network (DGCN), hyperspectral image (HSI) classification, superpixel segmentation.**

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I. Introduction

HYPERSPECTRAL image (HSI) is a technology that captures fine spectral characteristics of objects through hundreds of continuous narrowbands. This enables subpixel-level material identification, which possesses value for application in areas such as geological surveys and precision agriculture [1], [2], [3]. In contrast to conventional remote sensing data, the spectral–spatial integration characteristic of HSI poses dual challenges [4], [5]: continuous spectral evolution in the spectral dimension and microscopic spatial variation in the spatial dimension. On the one hand, nonlinear spectral changes exist between adjacent bands due to atmospheric scattering and mixed pixel effects. Furthermore, influenced by factors such as lighting and climatic conditions, similar objects exhibit spectral heterogeneity at different spatial locations.

Deep learning models have demonstrated significant advantages through the automatic extraction and selection of features via end-to-end training. Nevertheless, extant deep learning methodologies continue to exhibit manifest limitations. While proficient in spatial feature extraction, 2D-CNN often compromise the sequential integrity of spectral information [7]. Although 3D-CNN are theoretically capable of modeling both spectral and spatial features simultaneously, they remain constrained by the fixed size of the local receptive field and the cubic increase in computational complexity [8], which substantially hinders their ability to process large-scale remote sensing images.

Transformer effectively models the global dependencies of spectral sequences through self attention mechanism, overcomes the local receptive field limitation of traditional convolution operations, and significantly improves the capture efficiency of long-distance spectral features with its parallelization sequence processing capability. However, the application of this architecture in the field of hyperspectral imaging still faces fundamental challenges: its computational complexity increases by the square of the input sequence length,

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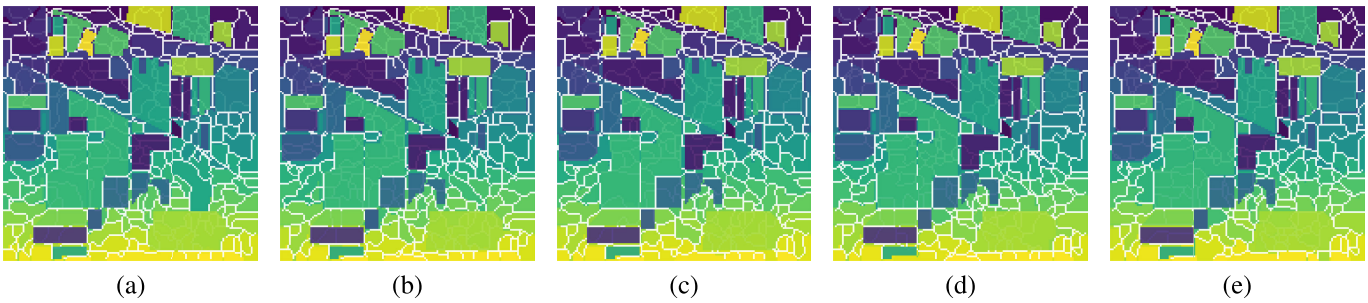


Fig. 1. Multitemporal superpixel boundaries overlaid on the Indian Pines ground truth. (a)–(e) Segmentation labels $L^{(0)}$ to $L^{(4)}$, illustrating the temporal evolution of superpixel structures.

resulting in huge computational overhead when processing high-dimensional spectral data. Meanwhile, although positional encoding can preserve sequence order, it lacks explicit modeling of local spatial structures, which can lead to feature distortion in areas with complex textures [9]. Although subsequent studies have proposed spectral–spatial, dual-branch, or local–global hybrid attention mechanisms to alleviate the above problems [24], their high dependence on large-scale annotated data and the reality of scarce hyperspectral samples still constrain the model's generalization ability and practical boundaries.

GCN constructs graph structures with spectral–spatial representations, leveraging neighborhood feature aggregation to model global spatial patterns and effectively capture implicit spatial dependencies between nodes [11]. GCN models based on superpixels can uniformly represent local and global features [12]. However, these models are subject to strong constraints imposed by the spectral–spatial uniformity assumption [13]. This alters the attribution of pixels at boundaries of heterogeneous objects, thereby inducing two types of feature pathologies: topological attribution mismatch in the spatial dimension and homogenization of statistical measures in the feature dimension [14], [15]. These pathologies ultimately lead to the extinction of native spectral features and the failure to analyze true object attributes. Whilst the CNN–GCN hybrid architecture has been demonstrated to be effective in the extraction of local features within superpixels and their global correlations [16], it is evident that the architecture is still susceptible to degradation of object separability. This is due to a lack of nonlinear spectral response modeling, and the lack of robust mechanisms for handling abnormal nodes results in the propagation of topological noise.

Dynamic graph convolutional networks (DGCN) address the topology mismatch problem by dynamically adjusting heterogeneous edge spectral–spatial correlation learning. However, existing methods are still limited to edge dynamic mechanisms, indirectly optimizing node features only by updating the weights of the adjacency matrix [17]. Essentially, they maintain a shallow optimization model where nodes are fixed, and edges change, forcing pixels to be embedded in a static and discrete topological framework, thereby depriving them of their expressive power in multigraph spaces [18]. To break through this limitation and construct a more physically reasonable representation, this article maps the spectral dimension to a virtual time series, corresponding to the inherent physical evolution process of ground spectral features in real scenes: the spectral characteristics of substances exhibit continuous and gradual changes with wavelength. The mapped virtual sequence accurately simulates the evolution trajectory of the spectral characteristics of ground objects under varying biochemical conditions. This modeling approach naturally captures the inherent dependencies and contextual transitions of the spectral dimension, allowing pixel representations to break through static topological constraints in structured relationship evolution, thereby enhancing physical interpretability and discriminative power in multigraph spaces.

The overlay diagram of superpixel segmentation results versus true object labels across each band group in Fig. 1 indicates that while the superpixel segmentation method effectively delineates homogeneous regions, it exhibits shortcomings in boundary precision and fragmented areas. Its misclassifications may lead to structural errors in the constructed map, thereby causing feature learning bias.

This article proposes the dynamic adaptive graph convolutional network (DAGCN), an efficient solution that addresses the core challenges of graph fusion. Its architecture integrates the dynamic graph sequence (DGS) construction strategy, multiscale adaptive graph convolution (MAGC), and spatio-temporal graph collaborative fusion (STGC). Specifically, we employ DGS to construct dynamic graph sequences, enabling differentiated representations of HSI. Subsequently, the multiscale strategy in MAGC extracts multiscale feature information from high-resolution spatial images. Concurrently, adaptive convolution modules applied at each layer generate adjacency matrices and node feature matrices, effectively mitigating the mosaic pixel issue. STGC incorporates a dynamic segmentation-based graph update (DSG Update) module and a multispectral channel attention (MSCA) mechanism. DSG update module dynamically optimizes the graph structure based on the adjacency matrix from MAGC, thereby circumventing defects in graph construction, as illustrated in Fig. 1. MSCA introduces self-attention mechanisms for different temporal feature maps, effectively capturing discriminative features across multispectral bands.

The main contributions of this work are summarized as follows.

1) To resolve spectral–spatial decoupling and noise sensitivity, we propose the DGS strategy, which maps high-dimensional spectral slices to multigraph space through band slicing and superpixel segmentation,

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achieving the transformation from pixels to regional prototypes.

2) To address spatial adaptability and feature granularity limitations, we designed the MAGC model: employing region-aware convolutions to learn neighborhood information, utilizing hybrid pooling to capture multiscale micro-level variations, achieving scale alignment through upsampling, and preserving pixel details via residual connections.
3) To eliminate temporal–spatial feature drift and boundary misalignment, we propose the STGC strategy: DSG Update enhances multiimage topological continuity through morphological extraction and centroid redistribution, while integrating the MSCA module to select discriminative spectral band features, thereby achieving efficient classification.
4) Experiments demonstrate that on classification tasks across four publicly available HSI datasets, this method outperforms nine state-of-the-art approaches, achieving synergistic improvements in efficiency, accuracy, and robustness.

The remaining sections of this article are structured as follows. Section II reviewed the related work. Section III provides a comprehensive description of the proposed method. Section IV outlines the experimental results and analyses. Conclusion and future work are discussed in Section V.

II. Related Works

A. HSI Classification Based on CNN

The advent of the deep learning revolution has precipitated the emergence of CNN as the preeminent modality for feature extraction in the domain of HSI. The hierarchical architecture of these neurons progressively fuses spectral–spatial information through local receptive fields, significantly enhancing feature discriminability. In order to surmount the training impediments that are prevalent in deep networks, cross-hierarchical feature remapping mechanisms have been proposed. Notable amongst these is reverse feature injection, a technique designed to enhance gradient propagation robustness. In order to address the challenges posed by high-dimensional small samples, tensor regularization constraint methods have been integrated into the 3-D convolutional framework, thereby effectively suppressing overfitting.

The pioneering work of Hu et al. [19] introduced a weighted five-layer 1D-CNN operating directly on the spectral signatures of HSI, establishing the first use of CNNs for hyperspectral classification. To overcome this limitation of ignoring spatial context, 2D-CNNs were subsequently developed. For example, Liu et al. [20] designed a semi-supervised 2D-CNN that automatically captures the contextual features of target pixels, enabling integrated spatial–spectral feature learning. Building upon 2D-CNNs, Li et al. [21] further advanced the field by proposing a 3D-CNN architecture for the direct extraction of joint spatial–spectral features, making it a more natural and effective framework for the HSI classification task.

The incorporation of conventional modules into HSI data analysis has facilitated the creation of many models with notable capacity for feature representation. For instance, Song et al. [23] designed a lightweight network that integrates 2D-CNN and a weighted sparse attention mechanism to process hyperspectral feature information in independent paths through information redistribution and attention weighting. Lu et al. [22] introduced a multiscale network utilizing 3-D convolutions and spatial attention, which adopts a parallel 3-D residual design to strengthen the learning of multiscale characteristics.

B. HSI Classification Based on Transformer

Transformer models, leveraging their powerful self-attention mechanism, can effectively capture global spectral–spatial features in HSI, providing a robust tool for HSI classification [24], [25], [26].

To directly address long-range dependencies in the spectral dimension, Hong et al. [27] formulated the SF model, utilizing a spectral-level self-attention mechanism to effectively capture global contextual information. The challenge of combining low-level features with high-level semantics is tackled by Sun et al. [28] in SSFTT, which merges CNN and Transformer. Their method first extracts shallow features using convolutional layers, converts them via a Gaussian-weighted tagger, and finally employs a Transformer encoder for advanced semantic modeling, enhancing performance efficiently. For the problem of joint spatial–spectral modeling, Peng et al. [29] proposed the CASST model with a dual-branch structure to separately handle fine-grained spatial and spectral features. Its novel cross-attention module with weighted sharing efficiently extracts interactive features, ensuring consistency and lowering computational load. To exploit multiscale information in HSI data, Wei et al. [30] designed a Transformer variant featuring a multiscale extraction module and a grouped fusion strategy. This architecture effectively integrates features across scales to improve object discrimination, and its innovative fusion of group convolution with Transformer maintains global modeling power while significantly boosting sensitivity to local details. Building upon the pursuit of adaptive multiscale representation, Xu et al. [31] proposed the DSFormer model. It advances beyond fixed fusion strategies by introducing a dual selective fusion mechanism. This mechanism adaptively determines the optimal receptive fields for various objects and selectively concentrates attention on the most salient spectral–spatial tokens, thereby enhancing discriminative power while mitigating computational redundancy.

However, Transformer models still face challenges in HSI classification. High computational complexity remains a significant issue; although structural designs have attempted to reduce computational costs, computational efficiency requires further optimization for particularly large-scale HSI datasets. Second, Transformers demand substantial training data, yet labeled samples in HSI classification are often limited, potentially constraining their performance. Meanwhile, there exists a fundamental mismatch between its architecture derived from natural language processing and continuous spectral signals, making it difficult for standard self-attention to effectively

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model the physical constraints of spectra and potentially oversmoothing subtle differences. Ultimately, achieving a finer balance between capturing local detailed features and global contextual information, as well as more effectively integrating spectral and spatial dimensional data, remains an area in urgent need of exploration.

C. HSI Classification Based on GCN

Graph convolutional networks (GCNs) have become a dominant framework for HSI classification, primarily owing to their capacity for capturing intricate, non-Euclidean interactions among pixels. They operate by building a graph where nodes represent spectral entities and edges denote their spectral–spatial similarities. This formulation allows GCNs to assimilate both local contextual details and long-range dependencies within the data, thereby substantially augmenting the discriminative features available for classification.

A significant challenge in applying GCN to HSI is the efficient and expressive construction of the underlying graph structure. It is evident that a number of GCN-based HSI classification methods have been put forward currently [32], [33], [34]. Specifically, Ding et al. [35] developed a class-imbalanced graph convolutional smoothing method, leveraging GCN's global properties to achieve adaptive feature correction. For small-sample classification problems, Zuo et al. [36] employed a global–local feature adaptive fusion method. By constructing a dual-branch network architecture, they separately extracted global contextual features and local detail information from images. An adaptive weight fusion mechanism was designed, effectively enhancing classification performance under limited training samples. Zhao et al. [37] designed a graph Transformer network integrating meta-path attention with graph attention mechanisms, effectively enhancing feature representation capabilities for heterogeneous graph structures while strengthening feature propagation at key nodes. To address the challenges of mechanical fault diagnosis under time-varying speeds, Yu et al. [38] proposed a weighted fusion framework integrating CNN and GCN, where a fusion strategy adaptively combines the graph and long-range features extracted by the two networks, achieving robust diagnostic performance. Hu et al. [39] proposed a self-supervised graph mask autoencoder, which learns expressive multiscale spatial–spectral features from unlabeled data through the masking and reconstruction tasks of the graph, further reducing the dependence on scarce labels and enhancing feature extraction, exploring the self-supervised paradigm.

Inherent graph properties face limitations during graph convolution, making dynamic graph convolution strategies a key future development direction. In a pioneering paper, Wan et al. [40] proposed a multiscale dynamic graph based on superpixels. This method establishes multiple latent graphs at the same level through a multiconvolution kernel strategy, providing a novel framework for guiding edge evolution. Focusing on the trade-off between accuracy and efficiency, Liu et al. [41] proposed a fast DGCN and CNN parallel network (FDGC), which employs a dynamic graph convolution module to extract intrinsic structural information in parallel with CNN branches, significantly improving training and inference speed while maintaining high classification accuracy. As demonstrated by Liu et al. [42], employing a CNN–GCN parallel architecture combined with a trainable adjacency matrix enables dynamic feature aggregation, thereby improving computational efficiency and overcoming the limitations of traditional GCN static topologies.

III. Proposed Network

This section first introduces representative approaches to DGCN, then outlines the overall architecture of this article, and finally presents our research contributions across three dimensions: graph structure construction, node feature learning, and dynamic updating and fusion of multitemporal graphs.

A. Preliminaries

A dynamic graph is a graph structure with evolving topology or attributes over time. DGCNs are designed to learn graph representations by aggregating features from spatial and temporal neighborhoods. Stacking multiple convolutional layers enables the capture of high-order evolutionary dependencies between nodes, facilitating the extraction of temporally aware deep features. This approach has proven highly effective for tasks like dynamic node classification and temporal link prediction. Hyperspectral data demonstrates a sequential evolution of features across spectral bands, whose gradual nature aligns well with discrete-time dynamic graph modeling. This network effectively captures such spectral–spatial joint dynamics, with its core working principle described as follows.

For a given dynamic graph tuple $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T})$ in a discrete time, where $\mathcal{T}$ represents a sequence of discrete-time steps, $\mathcal{V}$ and $\mathcal{E}$ denote the sets of all time nodes and edges, respectively, and both are allowed to dynamically add, remove, or change. Each time snapshot is represented as $\mathcal{G}^{(t)} = (\mathcal{V}^{(t)}, \mathcal{E}^{(t)})$, whose topological structure is fully defined by the adjacency matrix $\mathbf{A}^{(t)}$, with size $N_t \times N_t$, where N_t is the number of nodes at time t. If there is an edge between node v_i and node v_j, then $\mathbf{A}_{ij}^{(t)} > 0$. Each node has an d_t-dimensional feature vector, and $\mathbf{X}^{(t)} \in \mathbb{R}^{N_t \times d_t}$ represents the initial feature matrix of all N_t nodes at time t.

The DGCN learns new node embeddings $\mathbf{H}^{(t)}$ by spatiotemporally aggregating adjacent nodes. For snapshot $\mathcal{G}^{(t)}$, the following is performed:

$$\mathbf{Z}^{(t)} = \sigma\left(\tilde{\mathbf{A}}^{(t)}\mathbf{X}^{(t)}\mathbf{W}_{\text{sp}}\right) \tag{1}$$

where $\mathbf{Z}^{(t)}$ denotes the intermediate embedding of N_t nodes at time t generated by graph convolution, $\tilde{\mathbf{A}}^{(t)}$ denotes the adjacency matrix with self-loops at time t, and $\mathbf{W}_{\text{sp}}^{(t)} \in \mathbb{R}^{d_t \times d'}$ denotes a learnable weight matrix for spatial feature transformation at time t, and $\mathbf{Z}^{(t)} \in \mathbb{R}^{N_t \times d'}$. Subsequently, the historical state is integrated through a recurrent unit

$$\mathbf{H}^{(t)} = \phi\left(\mathbf{Z}^{(t)}, \mathbf{H}^{(t-1)}; \mathbf{W}_{\text{tmp}}\right) \tag{2}$$

where $\mathbf{H}^{(t)}$ denotes the embedding of N_t nodes at time t, $\phi(\cdot)$ denotes the temporal model, and $\mathbf{W}_{\text{tmp}}$ denotes the temporal weight parameter.

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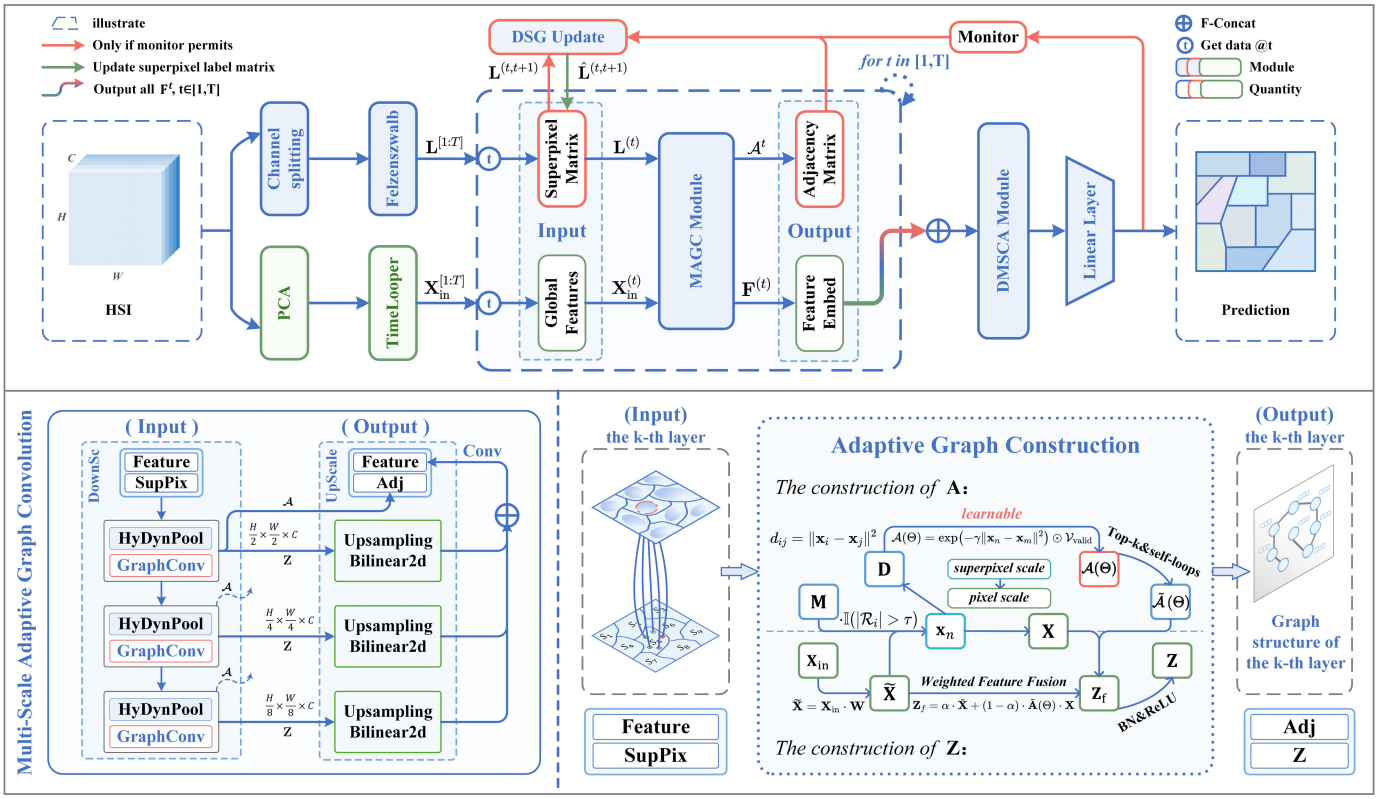


Fig. 2. Overall framework diagram of DAGCN. MAGC extracts multiscale feature information from HSI, utilizing adaptive convolutions to construct the graph structure and generate node feature matrices and adjacency matrices. DSG Update module dynamically updates the graph structure at each scale based on the adjacency matrix output by MAGC, thereby enhancing its feature representation capability. Furthermore, MSCA introduces independent attention mechanisms for feature maps across different temporal phases, effectively capturing discriminative features within distinct spectral band combinations.

B. Proposed DAGCN

Although existing DGCN-based methods enable dynamic learning and updating of graph features via adjacency matrices, HSI classification accuracy remains constrained by rigid graph skeletons. Therefore, we investigate dynamic graph update strategies and propose a pixel-level DAGCN model for HSI classification. First, the DGS module constructs multitemporal dynamic graph skeleton sequences, providing an adaptive topological foundation. Subsequently, the MAGC module learns from each temporal phase within these sequences to generate multiscale adjacency matrices and node features. Finally, the STGC strategy utilizes the full-resolution adjacency matrix produced by MAGC to execute dynamic graph structure updates and achieve effective multitemporal image fusion. These three modules operate sequentially and cooperatively, forming a cohesive framework that integrates dynamic topology construction, multiscale feature learning, and spatiotemporal information fusion. The overall model architecture is illustrated in Fig. 2, and its dynamic update mechanism is depicted in Fig. 3.

C. DGS Strategy

For large-scale HSIs, pixel-based graph construction methods can accurately characterize pixel spectral features, but they suffer from significant computational efficiency bottlenecks and noise interference issues. To address this, this article uses a superpixel cutting algorithm to aggregate spectro spatially homogeneous regions into graph nodes and uses regional mean features as node representations to achieve a feature transformation from the pixel level to the object level. This significantly reduces the scale of graph nodes through region merging, aggregating pixel-level nodes into superpixel-level nodes, reducing the number of nodes by 1 and 2 orders of magnitude, and significantly lowering the computational complexity of graph convolution. At the same time, it introduces spatial constraint mechanisms that effectively suppress salt-and-pepper noise in traditional pixel-level classification. Superpixel aggregation enhances the regional features of rare-sample categories by merging homogeneous regions, providing a more robust representation. This allows graph neural networks to optimize spectral and spatial features synergistically with lower computational power. Traditional methods rely on fixed features derived from PCA dimensionality reduction for segmentation, which makes it difficult to model the continuous dynamic response process of objects in the spectral domain. To address this issue, this article proposes a dynamic graph sequence design whose construction process is as follows.

Given the original hyperspectral data cube $\mathbf{I} \in \mathbb{R}^{H\times W\times C}$, where H and W represent the height and width, respectively, and C represents the spectral dimension. Then, the original data is uniformly divided into T subcubes along the spectral dimension by

$$\mathbf{I}^{(t)} = \mathbf{I}_{[:,:,(t-1)\cdot\Delta+1:t\cdot\Delta} \in \mathbb{R}^{H\times W\times\Delta} \tag{3}$$

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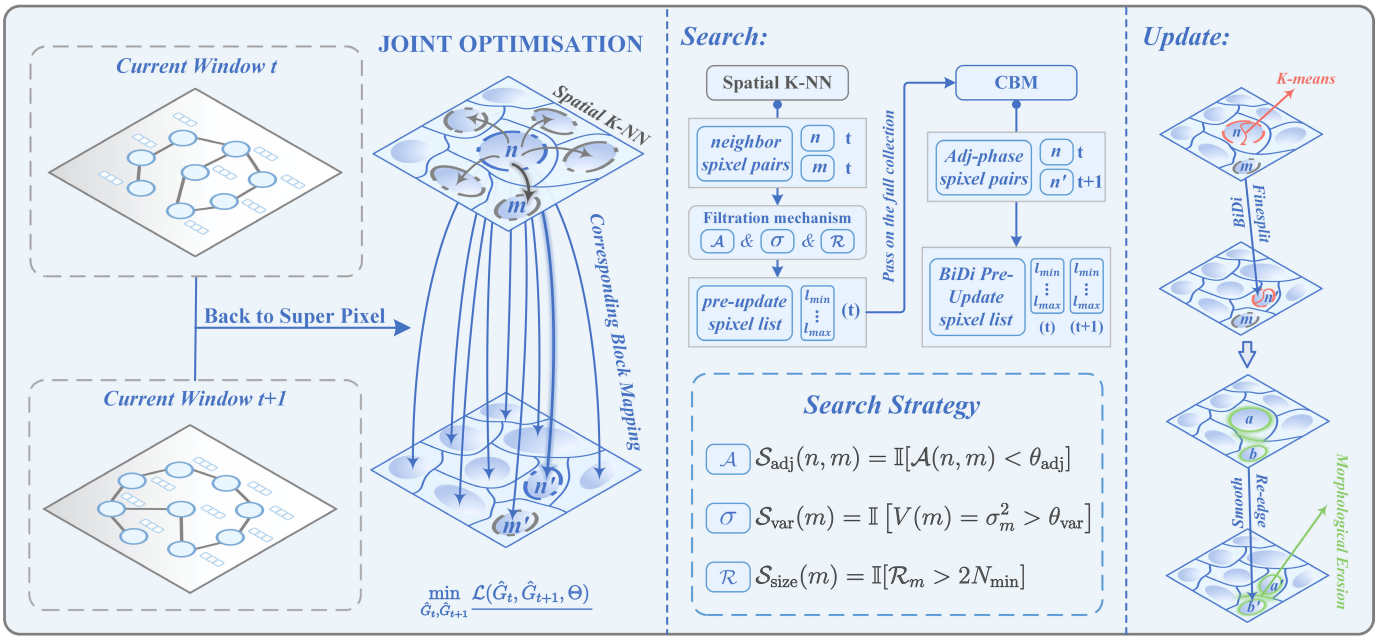


Fig. 3. Illustration of the DSG Update module. The search operation identifies adjacent temporal superpixel pairs to be updated, while the update operation performs bidirectional updating of these pairs through segmentation followed by edge smoothing.

where $\Delta = \lfloor C/T \rfloor$ is the subcube band number, and the mapping from the spectral dimension to the virtual time axis is successfully established as $\Phi : \{\mathbf{I}^{(t)}\}_{t=1}^{T} \to \{\tau_t\}_{t=1}^{T} \in \mathcal{T}$, τ_t denotes the tth timestamp.

Then, for each subcube $\mathbf{I}^{(t)}$ at each time step, the superpixel label matrix $L^{(t)}$ and node feature vectors are given by the Felzenszwalb algorithm

$$L^{(t)} = \mathcal{F}_{\text{Felz}}\left(\mathbf{I}^{(t)}; \sigma, \kappa, \min_{\text{size}}\right) \in \mathbb{Z}^{H\times W} \tag{4}$$

where σ is the standard deviation of the Gaussian smoothing kernel, κ is the merging threshold parameter, and $\min_{\text{size}}$ is the minimum region size.

To overcome the problem of node feature noise in multitemporal images, this article performs PCA dimension reduction on the entire HSI to obtain a d-dimensional feature matrix, and generates node features for each temporal phase based on the superpixel label matrix for each temporal phase

$$\mathbf{x}_n^t = \frac{1}{|\mathcal{R}_n^t|} \sum_{(i,j)\in\mathcal{R}_n^t} \mathbf{P}(i,j) \tag{5}$$

where $\mathbf{P}$ is the feature matrix of the global HSI after PCA dimension reduction. $\mathcal{R}_n^t = \{(i,j) \mid L^{(t)}(i,j) = n\}$ is the set of pixel coordinates of the nth superpixel S_n^t, l_n^t is the label value for the superpixel region S_n^t. At the same time, each node feature $\mathbf{x}_n^t$ will serve as the implicit representation of all pixels in the graph structure under its corresponding superpixel region.

In summary, we have defined the dynamic graph skeleton: $\mathcal{G}_{\text{skel}} = (\mathcal{V}, \mathcal{T})$, and established a bidirectional connection between the original image and the graph structure. The time domain $\mathcal{T}$ is established by spectral slice mapping, the vertex set $\mathcal{V}$ is constructed through spatiotemporal superpixel aggregation, and the edge set $\mathcal{E}$ is left undefined at this stage and will be constructed later by the MAGC module. The superpixel label matrix $L^{(t)}$ will serve as the spatial constraint prior for the subsequent DSG Update module.

D. MAGC Module

HSI classification tasks often face challenges stemming from the complexity of spatio–temporal–spectral feature coupling and significant intraclass variability, making pixel-level precise classification difficult to achieve. Traditional graph neural networks for image analysis typically rely on static learnable weight matrices W_{sp} for node feature transformation. This design suffers from two major flaws: over-enhanced prior assumptions and computational redundancy. Its complexity can be assessed through attention mechanisms. Furthermore, ground objects in HSIs exhibit diverse spatial scales, making it difficult for a single scale to simultaneously capture local details and global structures.

To address these challenges, we propose the MAGC module. This module achieves multiscale spatiotemporal spectral feature fusion through adaptive region-aware graph convolutions (RAGCs), as illustrated in the lower part of Fig. 2. The specific processing flow is as follows.

We use adaptive RAGCs to process the graph skeleton $\mathcal{G}_{\text{skel}}^{(t)} = (\mathcal{V}^t)$ at time t, dynamically aggregates spectral–spatial features of neighboring superpixel blocks based on graph structure $L^{(t)}$, and extracts multiscale contextual information using a mixed pooling mechanism. After K layers of RAGC iteration and feature fusion, the model can capture hierarchical spatial structure dependencies and spectral discriminative features. The kth layer of MAGC can be expressed as follows.

We transform the input features $\mathbf{X}_{\text{in}}^{(k)}$ through a linear projection and use the resulting feature matrix $\widetilde{\mathbf{X}}^{(k)}$ to refresh the

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graph node representations $\mathbf{x}^k$ by

$$\widetilde{\mathbf{X}}^{(k)} = \mathbf{X}_{\text{in}}^{(k)} \cdot \mathbf{W}^{(k)} \tag{6}$$

$$\mathbf{x}_n^k = \frac{1}{|\mathcal{R}_n^k|} \sum_{(i,j)\in\mathcal{R}_n^k} \widetilde{\mathbf{X}}^{(k)}(i,j) \quad \in \mathbb{R}^d \tag{7}$$

where $\mathbf{W}^{(k)} \in \mathbb{R}^{d\times d}$ is a learnable weight matrix.

$$\mathcal{V}_{\text{valid}}^k = \left\{ n \in [1, N_k] \mid \|\mathcal{R}_n^k\| > \theta \right\}. \tag{8}$$

Then, through an adaptive mechanism and sparsification, we construct the graph structure as follows. The affinity between nodes n and m is computed as

$$\mathcal{A}_{nm}^{(k)} = \exp\left(-\gamma \|\mathbf{x}_n^k - \mathbf{x}_m^k\|^2\right) \cdot \mathbb{I}\left(n, m \in \mathcal{V}_{\text{valid}}^k\right) \tag{9}$$

where γ is an adaptive temperature parameter, and $\mathbb{I}(\cdot)$ is an indicator function that returns 1 if both nodes are valid. The adjacency matrix is then sparsified, and self-loops are added

$$\tilde{\mathcal{A}}^{(k)} = \text{Sparsify}\left(\mathcal{A}^{(k)}\right) + \mathbf{I}_{N_k} \tag{10}$$

where Sparsify$(\cdot)$ retains only the top-K largest values in each row of the input matrix (setting others to zero), and $\mathbf{I}_{N_k}$ is the $N_k \times N_k$ identity matrix. This reduces computational complexity from $\mathcal{O}(N_k^2)$ to $\mathcal{O}(N_k K)$.

To prevent excessive smoothing and preserve original pixel-level features, the pixel-level intermediate embedding $\mathbf{Z}^{(k)}$ is computed with a residual connection

$$\mathbf{Z}^{(k)} = \sigma\left(\text{Norm}\left(\alpha \cdot \widetilde{\mathbf{X}}^{(k)} + (1-\alpha) \cdot \underbrace{\tilde{\mathcal{A}}^{(k)} \cdot \mathbf{X}^{(k)}}_{\text{AGCN}}\right)\right) \tag{11}$$

where α is the residual weight, σ is an activation function, and Norm denotes the normalization operation, choosing BatchNorm for large-scale HSI and LayerNorm for small-scale HSI based on empirical performance. The graph convolution feature (AGCN term) is computed at the superpixel scale and then shared across all pixels within each superpixel block.

The mixed pooling strategy combines MaxPool's salient feature extraction capability with SoftPool's context preservation feature to achieve collaborative fusion and restoration of multiscale discriminative features and spatial detail information with the assistance of upsampling, as follows:

$$\mathbf{Z}_{\text{pool}}^{(k)} = \lambda \cdot \text{MaxPool}\left(\mathbf{Z}^{(k)}\right) + (1-\lambda) \cdot \text{SoftPool}\left(\mathbf{Z}^{(k)}\right) \tag{12}$$

$$\mathbf{F}^{(k)} = \text{Upsample}\left(\mathbf{Z}_{\text{pool}}^{(k)}\right). \tag{13}$$

Therefore, define the input graph for the next layer as

$$\begin{cases} L^{(k+1)} = \text{NN} - \text{up}\left(L^{(k)}\right) \\ \mathbf{X}_{\text{in}}^{(k+1)} = \mathbf{Z}_{\text{pool}}^{(k)}. \end{cases} \tag{14}$$

By referencing the original superpixel construction strategy and performing computations based on RAGC, feature sequences are obtained and fused. The final output representation is generated through a unified RAGC

$$\mathbf{F}_{\text{fused}} = \text{Concat}\left(\mathbf{F}^{(1)}, \mathbf{F}^{(2)}, \ldots, \mathbf{F}^{(K)}\right) \tag{15}$$

$$\mathbf{F}_{\text{out}} = \text{GC}\left(\mathbf{F}_{\text{fused}}\right). \tag{16}$$

Here, GC$(\cdot)$ is a RAGC layer that transforms concatenated features of dimension $\mathbb{R}^{K\times d}$ into the final representation $\mathbf{F}_{\text{out}} \in \mathbb{R}^{C_{\text{out}}}$, with an output dimension of d_{out}.

For each time phase t, the adaptive adjacency matrix generated when k = 1 is defined as the adjacency matrix $A^{(t)}$ at time t, thereby completing the definition of spatial edges and refining the dynamic graph skeleton into a complete dynamic graph structure $\mathcal{G}^{(t)} = (\mathcal{V}^{(t)}, \mathcal{E}^{(t)})$. At the same time, the output $\mathcal{F}_{\text{out}} = \{\mathbf{F}_{\text{out}}^{(1)}, \mathbf{F}_{\text{out}}^{(2)}, \ldots, \mathbf{F}_{\text{out}}^{(T)}\}$ of MAGC on the entire time axis will also serve as an important basis for HSI classification.

E. STGC Strategy

Multitemporal HSI data processing faces fundamental challenges in superpixel-level analysis: temporal matching misalignment caused by spatial dislocation of superpixel boundaries between adjacent time phases, feature representation ambiguity induced by superpixel size heterogeneity, and feature fusion distortion resulting from high spectral heterogeneity among neighboring superpixels. Traditional superpixel processing methods typically rely on fixed thresholds or static graph structures, which struggle to adapt to dynamic multitemporal scenarios and lack coordinated optimization of spatial-feature consistency.

To address these challenges, this article proposes the STGC strategy, comprising two core components: DSG Update model and MSCA model. DSG Update achieves adaptive evolution of superpixel structures through a segmentation mechanism based on spatial spectral adjacency, cross-temporal consistency constraints, and feature-aware edge optimization. MSCA assigns independent DCT attention to the feature matrix of each time phase to efficiently capture the discriminative representations of each phase.

DSG Update establishes cross-temporal superpixel mapping relationships to maintain structural consistency across the temporal dimension. For graph $\mathcal{G}^{(t)}$

$$\mathcal{M}(n)^{(t)} = \underset{n'^{(t+1)}}{\text{argmax}} \left|\mathcal{R}_n^{(t)} \cap \mathcal{R}_{n'}^{(t+1)}\right| \tag{17}$$

where $\mathcal{M}(n)^{(t)}$ denotes the corresponding mapping of superpixel n at timestamp t to timestamp $t+1$, while $\mathcal{R}_n^{(t)}$ and $\mathcal{R}_{n'}^{(t+1)}$ represent the superpixel regions at the two timestamps, respectively.

Based on this mapping, collaborative segmentation is achieved to ensure spatiotemporal continuity throughout the superpixel evolution process.

By integrating the triple threshold criteria of adjacency matrix similarity, internal feature dispersion, and superpixel block size, superpixel blocks requiring correction are screened to form set S

$$S = \left\{ n \mid \mathcal{A}\left(\Theta^1\right)_{nm}^{(t)} < \theta_{\text{adj}} \wedge \sigma_n^2 > \theta_{\text{var}} \wedge |\mathcal{R}_n| > \theta_{\text{size}} \right\}. \tag{18}$$

For each superpixel region $\mathcal{R}_n$ in S, employ segmentation based on a clustering algorithm

$$\mathcal{R}_{n,\text{split}} = \underset{\mathcal{C}\in\{\mathcal{C}_1,\mathcal{C}_2,\mathcal{C}_3,\mathcal{C}_4\}}{\text{argmin}} \sum_{\mathbf{x}\in\mathcal{C}} \left\|\mathbf{x}_p^t - \mu_{\mathcal{C}}\right\|^2 \tag{19}$$

where $\rfloor$ represents the superpixel block to which the pixel p after clustering partition belongs, $\{\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3, \mathcal{C}_4\}$ is the

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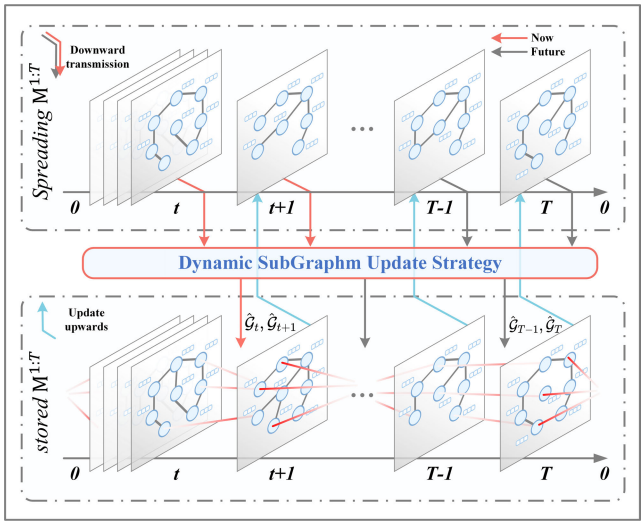


Fig. 4. Schematic of full-time phase diagram structure update logic.

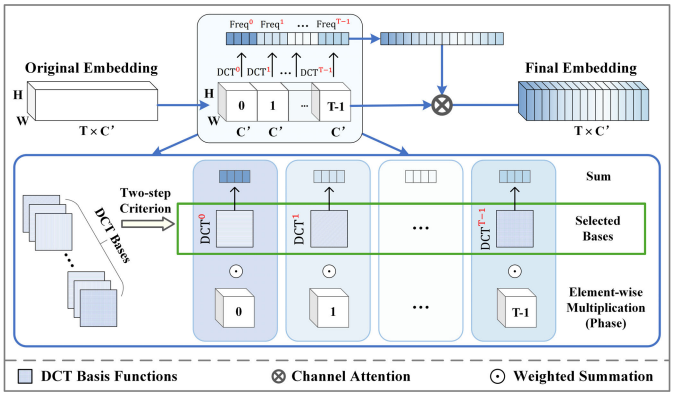


Fig. 5. Schematic of MSCA module operation.

superpixel block of the upper neighbor, lower neighbor, left neighbor and right neighbor of pixel p, $\mu_{\mathcal{C}}$ represents the feature center vectors for each subregion, and $\mathbf{x}_p^t$ denotes the feature vector for pixels within a superpixel.

To address boundary inconsistencies arising from segmentation, a feature-driven edge refinement method is employed. For a pixel p and the set of superpixels spatially adjacent to it, the process is achieved by calculating the similarity between p and the feature centers of its adjacent superpixels. The similarity between pixel p and an adjacent superpixel $\mathcal{R}_l$ is calculated as

$$s(p, \mathcal{R}_l) = \exp\left(-\gamma \left\|\mathbf{f}(p) - \mu_{\mathcal{R}_l}\right\|^2\right) \tag{20}$$

where l is the label of a superpixel adjacent to p, and $l \in \mathcal{N}(p)$. Here, $\mathcal{N}(p)$ denotes the set of superpixels adjacent to pixel p, $\mathbf{f}(p)$ is the feature vector of pixel p, and $\mu_{\mathcal{R}_l}$ is the feature center vector of superpixel $\mathcal{R}_l$.

Subsequently, the pixel p is reassigned to the superpixel with the highest similarity

$$\hat{\mathcal{R}}(p) = \underset{\mathcal{R}_l \in \mathcal{N}(p)}{\operatorname{argmax}}\, s(p, \mathcal{R}_l) \tag{21}$$

where $\hat{\mathcal{R}}(p)$ denotes the new superpixel to which pixel p is assigned. If $\mathcal{R}_n^{(t)}$ changes, update its corresponding superpixel region $\mathcal{R}_{n'}^{(t)}$ at time $t+1$ based on $\mathcal{M}(n)^{(t)}$.

Apply the DSG update module to update the logical sequence of the graph structure across all stages, as shown in Fig. 4.

To control the optimization granularity, the system processes no more than per iteration cycle and triggers corrections only when the training cycle satisfies epoch mod $T_{\text{epoch}} = 0$, where T_{epoch} denotes the superpixel correction cycle.

To capture cross-temporal spectral heterogeneity, we introduce the MSCA module, which incorporates a 2-D discrete cosine transform (DCT) for frequency-domain aggregation. This design assigns orthogonal frequency bases to each spectral phase, enhancing discriminative feature extraction and promoting cross-band interaction through frequency-adaptive feature recalibration. The frequency-domain strategy offers advantages in modeling intrinsic spectral correlations, as it compactly represents interchannel relationships via orthogonal frequency components, thereby emphasizing informative bands and suppressing redundant ones. This spectral-frequency mechanism complements the spatial–spectral features learned in preceding modules—which primarily focus on topological and spatial relations—by incorporating a complementary, frequency-aware recalibration of channelwise importance. The structure of this process is illustrated in Fig. 5.

We concatenate all the MAGC outputs of the multitemporal graph $\mathcal{G}$ to generate $\mathbf{F}_{\text{in}}$ as the input feature for the DCT ad follows:

$$\mathbf{F}_{\text{in}} = \text{Concat}\left(\mathbf{F}_{\text{out}}^{(1)}, \mathbf{F}_{\text{out}}^{(2)}, \ldots, \mathbf{F}_{\text{out}}^{(T)}\right). \tag{22}$$

Given an input feature tensor $\mathbf{F}_{\text{in}} \in \mathbb{R}^{C \times H \times W}$, where $C = C_{\text{out}} \times T$ represents the total number of channels obtained by concatenating multitemporal features (T denotes the number of temporal phases), this mechanism utilizes a predefined 2-D DCT weight matrix $\mathbf{W}_{\text{DCT}} \in \mathbb{R}^{C \times H \times W}$ to transform spatial features into the frequency domain. The matrix is governed by specified frequency indices fidx and fidx, with each element generated by the following orthogonal basis function:

$$a = \frac{\pi u(c)(w + 0.5)}{W}, \quad b = \frac{\pi v(c)(h + 0.5)}{H} \tag{23}$$

$$\mathbf{W}_{\text{DCT}}(c, h, w) = \cos(a)\cos(b) \tag{24}$$

where $u(c)$ and $v(c)$ denote the horizontal and vertical frequency components, assigned to channel c, respectively.

By performing weighted aggregation on the input feature maps, a frequency-domain aggregated feature vector can be obtained. For each channel c

$$\mathbf{z}_c = \sum_{h=1}^{H} \sum_{w=1}^{W} \mathbf{F}_{\text{in}}[c, h, w] \cdot \mathbf{W}_{\text{DCT}}[c, h, w] \tag{25}$$

where $\mathbf{z} \in \mathbb{r}^c$ is the frequency aggregation vector.

Feature recalibration is performed through a fully connected network with a bottleneck structure, learning independent attention weights for features at each temporal stage

$$\mathbf{A} = \sigma(\mathbf{W}_2 \cdot \delta(\mathbf{W}_1 \cdot \mathbf{z})) \tag{26}$$

where $\mathbf{W}_1 \in \mathbb{R}^{C/r \times C}$ and $\mathbf{W}_2 \in \mathbb{R}^{C \times C/r}$ are learnable weight matrices, and r is the compression ratio. The final output is the attention-weighted feature map

$$\mathbf{F}_{\text{final}} = \mathbf{X} \odot \mathbf{A}. \tag{27}$$

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TABLE I
CATEGORY NAME, SAMPLE NUMBERS OF TRAIN SET AND TEST SET OF EACH CLASS ON THE INDIAN PINES, PAVIA UNIVERSITY, HOUSTON 2013, AND WHU-HI-HANCHUAN DATASETS

	Indian Pines			Pavia University			Houston 2013			WHU-Hi-HanChuan		
No.	Category	Train	Test	Category	Train	Test	Category	Train	Test	Category	Train	Test
1	Alfalfa	5	41	Asphalt	332	6299	Healthy grass	126	1125	Strawberry	2237	42498
2	Corn-notill	143	1285	Meadows	932	17717	Stressed grass	126	1128	Cowpea	1138	21615
3	Corn-mintill	83	747	Gravel	105	1994	Synthetic grass	70	627	Soybean	515	9772
4	Corn	24	213	Trees	153	2911	Trees	125	1119	Sorghum	268	5085
5	Grass-pasture	48	435	Painted metal sheets	67	1278	Soil	125	1117	Water spinach	60	1140
6	Grass-trees	73	657	Bare soil	251	4778	Water	33	292	Watermelon	227	4306
7	Grass-pasture-mowed	3	25	Bitumen	67	1263	Residential	128	1140	Greens	296	5607
8	Hay-windrowed	48	430	Self-blocking bricks	184	3498	Commercial	125	1119	Trees	899	17079
9	Oats	2	18	Shadows	47	900	Road	126	1126	Grass	474	8995
10	Soybean-notill	97	875				Highway	124	1103	Red roof	526	9990
11	Soybean-mintill	245	2210				Railway	125	1110	Gray roof	846	16065
12	Soybean-clean	59	534				Parking Lot 1	125	1108	Plastic	184	3495
13	Wheat	20	185				Parking Lot 2	48	421	Bare soil	456	8660
14	Woods	126	1139				Tennis Court	43	385	Road	928	17632
15	Buildings	39	347				Running Track	67	593	Bright object	57	1079
16	Stone	9	84							Water	3771	71630
	Total	1024	9225	Total	2138	40638	Total	1503	13526	Total	11882	225757

Then, $\mathbf{F}_{\text{final}}$ is fed into the classifier for final classification as follows:

$$\text{pred} = \text{Classifier}\left(\mathbf{F}_{\text{final}}\right) \tag{28}$$

where Classifier(·) comprises a single linear layer with an output dimension equal to the number of categories. Classification is optimized using the cross-entropy loss function. The experimental section provides a comprehensive evaluation of the performance across different architectures.

IV. PERFORMANCE EVALUATION

A. Datasets Description

To evaluate the classification performance of the proposed DAGC, we conducted extensive comparative experiments on four publicly available HSI datasets: Indian Pines, Pavia University, Houston 2013, and WHU-Hi-HanChuan. Table I details the splitting of training and testing sets.

1) Indian Pines: This dataset was captured by the Airborne Visible and Infrared Imaging Spectrometer (AVIRIS) sensor in 1992. The flight campaign covered the Indian Pines study area, situated in Northwestern Indiana. The spectrometer's wavelength range spans 0.4 to 2.5 μm. By excluding bands covering water absorption regions, the number of spectral bands was reduced to 200. Spatial resolution is 20 m per pixel, covering an area of 145 $\times$ 145 pixels. The dataset includes 10 249 ground sampling points representing 16 different categories.

2) University of Pavia: This dataset was collected by the reflective optical system imaging spectrometer (ROSIS-3) sensor over the city of Pavia, Italy. The spectrometer's imaging wavelength range spans 0.43 to 0.86 μm. After excluding 12 noisy bands, 103 spectral bands were retained with a spatial resolution of 1.3 m. Image dimensions are 610 $\times$ 340 pixels, comprising 42 776 annotated samples across nine land cover categories.

3) Houston 2013: This dataset was acquired by the ITRES CASI-1500 airborne sensor over the University of Houston campus and surrounding areas, sourced from the 2013 GRSS data fusion. The imaging spectral range spans 0.38–1.05 μm, comprising 144 bands. Image dimensions are 340 $\times$ 1905 pixels with a spatial resolution of 2.5 m/pixel. The dataset contains 15 029 samples classified into 15 challenging land cover categories.

4) WHU-HI-HanChuan: This dataset was collected by a research team from Wuhan University and the Chinese Academy of Sciences using an airborne hyperspectral sensor in the Hanchuan region. The imaging spectral range spans 0.4–1.0 μm, comprising 274 spectral bands. Image dimensions are 1217 $\times$ 303 pixels with a spatial resolution of approximately 0.11 m/pixel. It contains 155 010 annotated samples covering 16 typical land cover categories.

B. Experimental Settings

1) Evaluation Metrics: Based on existing HSI classification methods, evaluation metrics include overall accuracy (OA), average accuracy (AA), and Kappa coefficient ($\mathcal{K}$). All experiments were conducted under identical experimental conditions, with reported results averaged across five consecutive trials.

2) Implementation Details: All experiments were conducted on the PyTorch platform using a single RTX 3090Ti GPU. Parameter updates employed the Adam optimizer with a learning rate of 0.01. Training cycles were set to 200 iterations. For datasets with fewer than 200 bands, we divided them into three time phases; those with 200 or more bands were divided into five time phases.

3) Competitive Approach: A comprehensive comparison was conducted among four representative HSI classification architectures, including conventional methods (SVM [6]), CNN-based methods (2D-CNN [20] and 3D-CNN [21]), Transformer-based methods (SSFTT [28], CASST [29], and DSFormer [31]), and GCN-based methods (FDGC [41], SGMAE [39], and WFCG [38]).

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TABLE II

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, $\mathcal{K}$, AND THE ACCURACIES OF EACH CLASS FOR THE INDIAN PINES DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional	CNN-based Methods		Transformer-based Methods			GCN-based Methods			Our Work
	SVM	2D-CNN	3D-CNN	SSFTT	CASST	DSFormer	FDGC	SGMAE	WFCG	DAGCN
1	33.17 ± 10.24	86.09 ± 2.42	83.91 ± 2.22	99.27 ± 2.20	92.93 ± 10.66	94.63 ± 4.47	**100.00 ± 0.00**	97.83 ± 5.24	96.10 ± 3.12	**100.00 ± 0.00**
2	73.00 ± 3.51	87.32 ± 0.74	83.67 ± 2.34	98.27 ± 1.07	97.67 ± 0.85	95.99 ± 1.28	96.20 ± 0.14	97.02 ± 0.45	98.26 ± 0.85	**99.22 ± 0.20**
3	60.08 ± 3.65	82.61 ± 1.55	78.57 ± 1.65	96.43 ± 2.03	98.77 ± 0.76	98.97 ± 1.09	97.52 ± 0.04	99.69 ± 0.17	98.73 ± 1.09	**99.14 ± 0.49**
4	49.77 ± 9.11	78.19 ± 4.07	83.63 ± 4.01	97.79 ± 1.85	99.86 ± 0.42	97.28 ± 2.42	**100.00 ± 0.00**	99.92 ± 0.17	99.48 ± 0.90	99.86 ± 0.30
5	87.06 ± 3.94	90.23 ± 0.90	88.47 ± 2.52	99.70 ± 0.45	98.39 ± 2.07	95.22 ± 1.89	99.71 ± 0.08	99.67 ± 0.30	98.32 ± 1.76	**99.77 ± 0.23**
6	95.60 ± 1.10	95.67 ± 0.92	93.44 ± 1.36	97.49 ± 2.07	98.42 ± 1.42	99.35 ± 0.45	99.17 ± 0.10	99.21 ± 0.75	99.73 ± 0.27	**100.00 ± 0.00**
7	62.40 ± 21.56	86.79 ± 6.97	67.86 ± 10.10	**100.0 ± 0.00**	80.00 ± 19.43	90.40 ± 11.34	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	98.80 ± 1.83
8	98.02 ± 1.09	97.78 ± 0.84	98.93 ± 0.66	**100.0 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	99.73 ± 0.09	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
9	32.78 ± 13.93	37.50 ± 2.50	35.50 ± 4.72	93.33 ± 16.44	72.78 ± 20.71	76.11 ± 16.11	**100.00 ± 0.00**	95.00 ± 10.49	97.22 ± 5.69	**100.00 ± 0.00**
10	74.74 ± 3.40	90.40 ± 0.46	88.07 ± 1.45	97.43 ± 1.21	96.01 ± 0.71	96.30 ± 2.57	97.29 ± 0.23	98.33 ± 1.58	98.34 ± 1.49	**99.73 ± 0.19**
11	85.21 ± 2.36	94.68 ± 0.50	91.88 ± 0.63	98.18 ± 0.56	99.33 ± 0.51	97.96 ± 1.02	92.83 ± 0.01	99.50 ± 0.14	99.18 ± 0.49	**99.44 ± 0.16**
12	65.79 ± 3.74	80.02 ± 0.95	81.47 ± 1.86	93.58 ± 2.88	96.33 ± 0.59	95.99 ± 1.33	95.38 ± 0.73	97.86 ± 1.51	98.72 ± 0.57	**99.55 ± 0.19**
13	95.51 ± 2.78	98.05 ± 0.44	98.05 ± 1.07	96.97 ± 1.33	99.62 ± 1.13	99.19 ± 0.88	**100.00 ± 0.00**	98.63 ± 0.89	99.67 ± 0.43	99.95 ± 0.16
14	96.68 ± 0.95	97.48 ± 0.22	97.87 ± 0.54	99.63 ± 0.31	99.42 ± 0.24	99.44 ± 0.33	99.72 ± 0.03	99.88 ± 0.15	99.95 ± 0.13	**100.00 ± 0.00**
15	53.75 ± 5.26	91.71 ± 1.36	82.12 ± 4.46	97.90 ± 0.98	98.24 ± 1.11	98.21 ± 3.70	99.65 ± 0.00	99.84 ± 0.17	99.51 ± 0.83	**100.00 ± 0.00**
16	84.52 ± 3.32	89.89 ± 2.64	83.23 ± 4.62	82.74 ± 5.30	61.43 ± 8.59	95.71 ± 3.92	**100.00 ± 0.00**	93.44 ± 2.12	99.28 ± 0.80	**100.00 ± 0.00**
OA (%)	80.00 ± 0.53	91.13 ± 0.24	88.99 ± 0.52	97.81 ± 0.53	98.00 ± 0.23	97.66 ± 0.34	96.63 ± 0.03	98.96 ± 0.33	99.06 ± 0.24	**99.62 ± 0.08**
AA (%)	71.76 ± 2.37	86.53 ± 0.62	83.54 ± 0.85	96.79 ± 1.38	93.08 ± 2.22	95.67 ± 0.91	98.58 ± 0.03	98.49 ± 1.13	98.91 ± 0.43	**99.72 ± 0.13**
Kappa	77.05 ± 0.62	89.86 ± 0.28	87.42 ± 0.60	97.51 ± 0.60	97.72 ± 0.26	97.34 ± 0.39	96.10 ± 0.04	98.81 ± 0.38	98.92 ± 0.28	**99.56 ± 0.09**

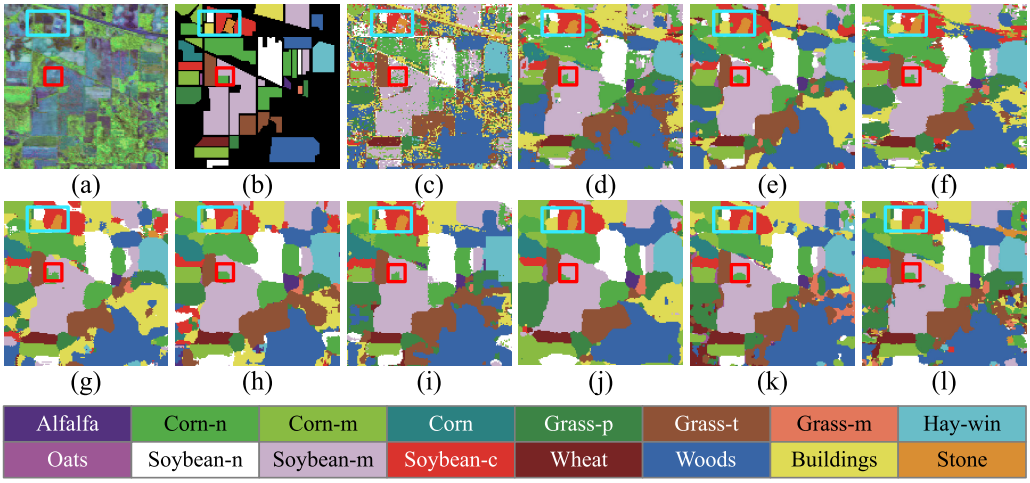


Fig. 6. Classification maps obtained by the compared methods on the Indian Pines dataset. (a) False-color image. (b) Ground truth. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) SSFTT. (g) CASST. (h) DSFormer. (i) FDGC. (j) SGMAE. (k) WFCG. (l) DAGCN.

C. Comparison With Competing Methods

Quantitative accuracy metrics (including OA, AA, and $\mathcal{K}$) achieved on the Indian Pines, Pavia University, Houston 2013, and WHU-Hi-HanChuan datasets are summarized in Tables II–V, with optimal results highlighted in bold. Corresponding classification visualization results are presented in Figs. 6–9.

1) Results on Indian Pines: The classification accuracy of different methods on the Indian Pines dataset is shown in Table II. It can be observed that the proposed method outperforms all nine comparison methods in terms of OA, AA, and $\mathcal{K}$. Among the 16 land cover categories, this method achieves higher classification accuracy than other methods in 13 categories. Specifically, it achieves 100% accuracy for alfalfa, grass-trees, hay-windrowed, oats, woods, buildings, and stone categories, with good results for the remaining categories. The SVM method achieved an OA of 80.00%. Among the two CNN-based classification models (2D-CNN and 3D-CNN), 2D-CNN performed best with an OA of 91.13%. Among the three Transformer-based methods (SSFTT, CASST, and DSFormer), CASST demonstrated the highest performance with an OA of 98.00%. Among the three GCN-based methods (FDGC, SGMAE, and WFCG), WFCG performed best with an OA of 99.06%. SGMAE achieved an OA of 98.96%, while our proposed DAGCN attained an OA of 99.62%, alongside an AA of 99.72% and a $\mathcal{K}$ of 99.56.

The classification results of different methods are shown in Fig. 6. For comparison, the figure presents the false-color images of the Indian Pines dataset and their ground-truth labels. Evidently, the SVM method performs the worst, exhibiting significant salt-and-pepper noise. The remaining methods with spatial features demonstrate superior noise suppression capabilities. Classification methods based on CNNs and Transformers lose some fine-grained features when leveraging spatial context information, as indicated by the blue-boxed regions. Within these areas, the DAGCN method maintains the best classification performance. Although GCN-based methods preserve feature details and contour characteristics, performance in edge regions remains suboptimal, as

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TABLE III

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, $\mathcal{K}$, AND THE ACCURACIES OF EACH CLASS FOR THE PAVIA UNIVERSITY DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional	CNN-based Methods		Transformer-based Methods			GCN-based Methods			Our Work
	SVM	2D-CNN	3D-CNN	SSFTT	CASST	DSformer	FDGC	SGMAE	WFCG	DAGCN
1	93.35 ± 0.48	96.10 ± 0.40	95.20 ± 0.72	99.70 ± 0.20	99.64 ± 0.21	99.96 ± 0.06	95.29 ± 0.16	99.59 ± 0.36	99.61 ± 0.40	**99.97 ± 0.05**
2	98.31 ± 0.14	99.32 ± 0.07	99.40 ± 0.12	99.95 ± 0.05	99.41 ± 0.35	99.92 ± 0.05	96.65 ± 0.26	99.93 ± 0.02	**99.99 ± 0.00**	99.99 ± 0.01
3	79.84 ± 0.98	89.29 ± 0.74	84.06 ± 1.96	97.22 ± 1.70	99.02 ± 0.41	98.43 ± 0.77	97.80 ± 0.15	99.04 ± 0.24	99.47 ± 0.40	**99.94 ± 0.05**
4	92.93 ± 0.81	95.86 ± 0.30	95.10 ± 0.91	93.50 ± 2.12	96.03 ± 1.50	98.52 ± 0.84	90.39 ± 0.32	98.01 ± 0.28	98.80 ± 0.88	**98.89 ± 0.27**
5	98.96 ± 0.37	98.45 ± 0.34	98.94 ± 0.59	99.12 ± 0.46	96.78 ± 1.52	99.98 ± 0.07	99.48 ± 0.12	97.13 ± 1.11	99.86 ± 0.28	**100.00 ± 0.00**
6	88.61 ± 0.89	98.46 ± 0.28	98.28 ± 0.50	99.59 ± 1.09	99.99 ± 0.01	99.91 ± 0.13	99.54 ± 0.03	99.98 ± 0.03	99.97 ± 0.00	**100.00 ± 0.00**
7	86.40 ± 1.23	77.65 ± 1.31	79.42 ± 3.51	99.77 ± 0.28	**100.00 ± 0.00**	99.88 ± 0.15	99.52 ± 0.02	**100.00 ± 0.00**	99.73 ± 0.22	99.90 ± 0.15
8	90.14 ± 0.76	96.13 ± 0.36	93.67 ± 1.40	99.34 ± 0.34	99.41 ± 0.41	99.08 ± 0.47	98.17 ± 0.03	99.63 ± 0.15	**99.66 ± 0.23**	99.60 ± 0.32
9	99.42 ± 0.26	94.19 ± 1.06	93.24 ± 1.46	93.31 ± 2.17	94.47 ± 0.19	98.57 ± 0.62	95.22 ± 0.18	96.29 ± 0.74	**99.87 ± 0.11**	98.68 ± 0.41
OA (%)	94.08 ± 0.16	96.89 ± 0.14	96.29 ± 0.41	99.04 ± 0.27	99.08 ± 0.18	99.65 ± 0.08	96.67 ± 0.12	99.51 ± 0.07	99.78 ± 0.10	**99.84 ± 0.04**
AA (%)	92.00 ± 0.23	93.94 ± 0.23	93.03 ± 0.79	97.94 ± 0.59	98.31 ± 0.25	99.38 ± 0.17	96.90 ± 0.07	98.85 ± 0.15	99.66 ± 0.14	**99.66 ± 0.06**
Kappa	92.12 ± 0.22	95.88 ± 0.19	95.08 ± 0.55	98.73 ± 0.35	98.78 ± 0.24	99.54 ± 0.10	95.68 ± 0.15	99.35 ± 0.09	99.70 ± 0.13	**99.79 ± 0.05**

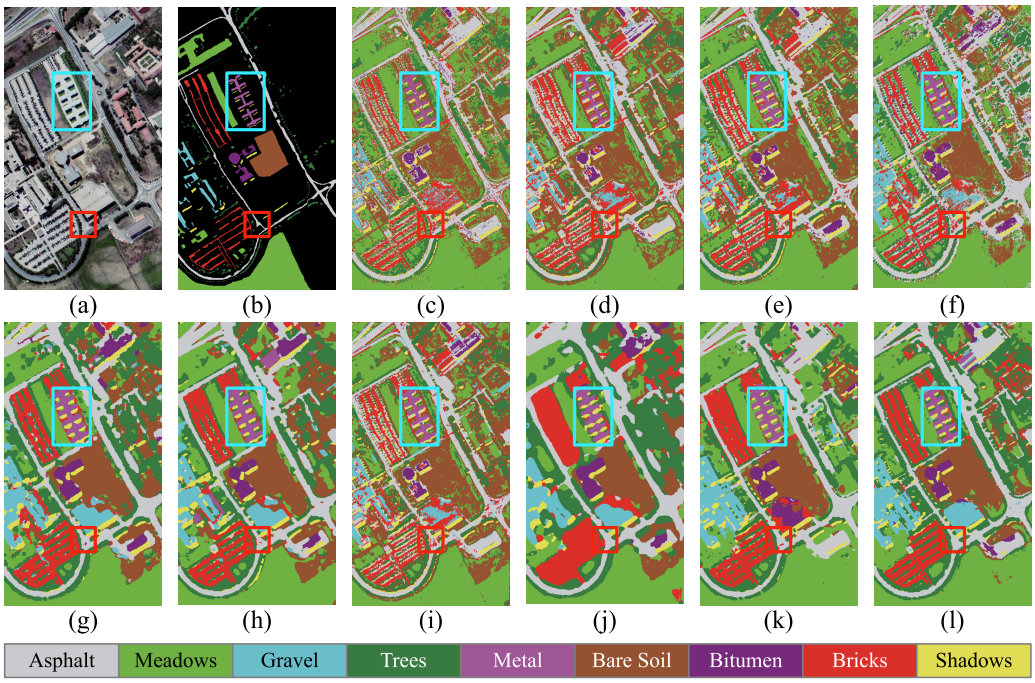


Fig. 7. Classification maps obtained by the compared methods on the Pavia University dataset. (a) False-color image. (b) Ground truth. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) SSFTT. (g) CASST. (h) DSFormer. (i) FDGC. (j) SGMAE. (k) WFCG. (l) DAGCN.

indicated by the red-boxed area in the figure. Within this region, the DSFormer method achieved the best classification results, but its same class block area is too large, resulting in lower classification accuracy in the blue box area. Overall, the DAGCN method demonstrates optimal performance, achieving high classification accuracy while maintaining clear boundary details and rich features.

2) Results on Pavia University: The classification accuracy of different methods on the Pavia University dataset is listed in Table III. It can be seen that this method achieves higher OA, AA, and $\mathcal{K}$ than the nine comparison methods. Among the nine feature categories, the classification accuracy for asphalt, gravel, trees, painted metal sheets, and bare soil surpasses that of other methods. Furthermore, the accuracy for the remaining categories differs only slightly from the best classification accuracy. The SVM achieves an OA of 94.08%. Among the two CNN classification models, 2D-CNN performed best with an OA of 96.89%. DSFormer was the optimal method among the three Transformer-based approaches, achieving an OA of 99.65%. For GCN-based classification methods, WFCG was the best with an OA of 99.78%, while SGMAE achieved an OA of 99.51%. Our proposed DAGCN model achieved the highest OA of 99.84%, alongside the highest AA score of 99.66% and the highest $\mathcal{K}$ of 99.79.

The classification results of different methods are shown in Fig. 7. For comparison, we present the Pavia University false-color imagery and the ground-truth label map. Evidently, the SVM classification yields the poorest results, with noticeable salt-and-pepper noise in its classified images, as well as in those produced by CNN-based classification methods. Transformer-based and GCN-based methods demonstrate superior classification performance in fragmented areas, particularly within the red and blue boxed regions. Among Transformer-based models, DSFormer achieves the best performance both within the boxed areas and overall. Among the three GCN-based classification models, WFCG exhibits the highest OA. In comparison, the DAGCN model proposed in this article not only achieves the highest test accuracy but also

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TABLE IV

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, $\mathcal{K}$, AND THE ACCURACIES OF EACH CLASS FOR THE HOUSTON 2013 DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional	CNN-based Methods		Transformer-based Methods			GCN-based Methods			Our Work
	SVM	2D-CNN	3D-CNN	SSFTT	CASST	DSFormer	FDGC	SGMAE	WFGC	DAGCN
1	97.23 ± 1.55	96.51 ± 4.20	96.25 ± 2.92	95.59 ± 3.96	95.58 ± 1.44	98.62 ± 1.69	93.17 ± 0.10	98.91 ± 0.63	99.48 ± 0.42	**99.83 ± 0.16**
2	98.41 ± 0.27	96.68 ± 1.43	94.04 ± 1.36	99.55 ± 0.56	99.47 ± 0.36	98.49 ± 2.14	94.77 ± 0.28	99.90 ± 0.07	99.45 ± 0.17	**99.95 ± 0.07**
3	99.52 ± 0.59	98.22 ± 0.87	99.10 ± 0.49	99.62 ± 0.84	99.87 ± 0.26	99.52 ± 0.47	99.30 ± 0.07	99.99 ± 0.04	99.97 ± 0.06	**100.00 ± 0.00**
4	98.23 ± 1.19	93.91 ± 3.01	94.72 ± 2.99	96.13 ± 4.47	95.07 ± 2.38	99.19 ± 1.12	87.67 ± 0.35	99.93 ± 0.12	96.18 ± 1.67	**99.91 ± 0.16**
5	97.71 ± 0.68	97.34 ± 3.64	98.95 ± 0.60	99.94 ± 0.16	99.93 ± 0.14	**100.00 ± 0.00**	97.55 ± 0.04	99.99 ± 0.02	**100.00 ± 0.00**	**100.00 ± 0.00**
6	94.11 ± 1.21	77.14 ± 2.27	79.91 ± 3.75	99.83 ± 0.51	**100.00 ± 0.00**	99.59 ± 0.50	**100.00 ± 0.00**	**100.00 ± 0.00**	96.16 ± 4.04	**100.00 ± 0.00**
7	89.89 ± 1.85	95.22 ± 2.60	92.80 ± 1.34	99.04 ± 0.85	96.86 ± 0.58	98.48 ± 1.09	91.64 ± 0.62	98.06 ± 0.68	98.88 ± 0.41	**99.91 ± 0.16**
8	90.96 ± 1.57	87.26 ± 3.21	85.88 ± 2.37	95.55 ± 0.91	**99.86 ± 0.15**	98.03 ± 1.36	95.39 ± 0.32	98.18 ± 0.78	96.34 ± 1.28	99.79 ± 0.20
9	87.45 ± 1.77	91.70 ± 4.56	90.96 ± 0.72	97.58 ± 1.24	98.26 ± 0.63	98.23 ± 1.16	90.17 ± 1.60	99.61 ± 0.52	95.88 ± 0.96	**98.67 ± 0.78**
10	92.93 ± 0.86	90.36 ± 10.98	93.51 ± 1.71	97.09 ± 4.58	**100.00 ± 0.00**	99.79 ± 0.27	99.95 ± 0.11	99.97 ± 0.07	99.49 ± 0.37	**100.00 ± 0.00**
11	87.52 ± 1.12	92.74 ± 2.71	91.32 ± 1.42	99.60 ± 0.87	99.93 ± 0.09	99.88 ± 0.19	**99.96 ± 0.07**	99.71 ± 0.44	99.28 ± 0.97	99.16 ± 0.75
12	86.79 ± 2.51	89.08 ± 11.14	91.41 ± 0.99	97.58 ± 0.87	99.68 ± 0.07	98.79 ± 0.84	97.58 ± 0.78	99.64 ± 0.06	99.48 ± 0.17	**99.50 ± 0.14**
13	34.31 ± 3.14	79.08 ± 3.76	62.67 ± 4.61	96.78 ± 2.26	96.82 ± 3.23	96.68 ± 2.26	98.04 ± 0.11	**98.81 ± 1.64**	96.73 ± 1.67	94.99 ± 0.25
14	98.60 ± 1.32	92.15 ± 3.40	89.81 ± 3.31	**100.00 ± 0.00**	**100.00 ± 0.00**	99.92 ± 0.17	**100.00 ± 0.00**	99.98 ± 0.07	**100.00 ± 0.00**	99.95 ± 0.16
15	98.01 ± 0.59	93.20 ± 6.22	96.06 ± 0.75	95.61 ± 6.19	96.56 ± 1.73	99.97 ± 0.07	99.32 ± 0.07	**100.00 ± 0.00**	**100.00 ± 0.00**	99.95 ± 0.08
OA (%)	91.64 ± 0.13	92.53 ± 3.41	92.08 ± 0.54	97.84 ± 0.98	98.46 ± 0.28	98.99 ± 0.25	95.44 ± 0.26	99.45 ± 0.12	98.52 ± 0.07	**99.57 ± 0.09**
AA (%)	90.11 ± 0.08	91.37 ± 2.85	90.49 ± 0.68	97.97 ± 0.89	98.53 ± 0.32	99.01 ± 0.22	90.28 ± 0.19	99.51 ± 0.15	98.49 ± 0.23	**99.44 ± 0.08**
Kappa	90.95 ± 0.14	91.92 ± 3.68	91.44 ± 0.58	97.66 ± 1.06	98.34 ± 0.30	98.91 ± 0.27	95.06 ± 0.28	99.41 ± 0.13	98.40 ± 0.08	**99.53 ± 0.10**

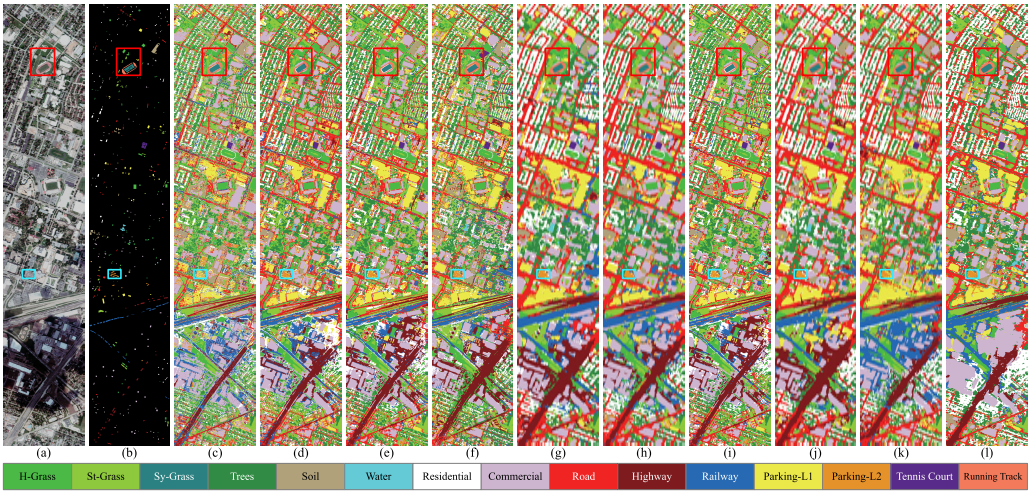


Fig. 8. Classification maps obtained by the compared methods on the Houston 2013 dataset. (a) False-color image. (b) Ground truth. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) SSFTT. (g) CASST. (h) DSFormer. (i) FDGC. (j) SGMAE. (k) WFCG. (l) DAGCN.

delivers the best classification results both at the boundaries and within the land cover areas, as demonstrated by the blue and red boxed regions.

3) Results on Houston 2013: The classification accuracy of different methods on the Houston 2013 dataset is shown in Table IV. It can be seen that the proposed method achieves higher OA, AA, and Kappa than all nine comparison methods. Among the 15 land cover classes, the classification accuracy for ten classes exceeds that of other methods, with synthetic grass, soil, water, and highway achieving 100% accuracy. The SVM method achieves an OA of 91.64%. Among the two CNN-based models, 2D-CNN performed best with an OA of 92.53%. Among the three Transformer-based methods, DSFormer achieved the highest performance with an OA of 98.99%. Among the three GCN-based methods, SGMAE demonstrated the best performance with an OA of 99.45%, closely followed by WFCG at 98.52%. Our proposed DAGCN achieves the optimal solution with an OA of 99.57%, AA of 99.44%, and a $\mathcal{K}$ of 99.53.

Fig. 8 displays the classification results of different methods. For comparison purposes, we simultaneously present the false-color image from the University of Pavia alongside the ground-truth label map. Clearly, SVM yields the poorest classification performance. The two CNN-based methods also exhibit similar noise characteristics, particularly evident in the red and blue highlighted regions. Methods based on Transformers and GCN demonstrate superior classification performance in fragmented areas. Among the Transformer models, CASST performs best in the red boxed region, while DSFormer achieves the highest OA. Among the three GCN-based classification models, SGMAE achieves the highest OA but misclassifies some isolated pixel clusters. In contrast, the proposed DAGCN model delivers optimal classification results in both fragmented and large contiguous areas.

4) Results on WHU-Hi-HanChuan: The classification accuracy of different methods on the Houston 2013 dataset is clearly shown in Table V. It can be seen that the proposed method outperforms all nine selected comparison methods

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TABLE V

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, $\mathcal{K}$, AND THE ACCURACIES OF EACH CLASS FOR THE WHU-HI-HANCHUAN DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional	CNN-based Methods		Transformer-based Methods			GCN-based Methods			Our Work
	SVM	2D-CNN	3D-CNN	SSFTT	CASST	DSformer	FDGC	SGMAE	WFCG	DAGCN
1	96.03 ± 0.38	86.09 ± 2.42	98.73 ± 0.15	99.54 ± 0.20	99.82 ± 0.09	99.22 ± 1.06	93.99 ± 0.09	99.69 ± 0.15	99.56 ± 0.20	**99.92 ± 0.11**
2	81.22 ± 0.82	87.32 ± 0.74	95.69 ± 0.79	98.98 ± 0.88	**99.80 ± 0.17**	99.65 ± 0.20	88.42 ± 0.07	99.00 ± 0.16	98.94 ± 0.48	99.59 ± 0.24
3	80.15 ± 0.49	82.61 ± 1.55	95.18 ± 1.96	99.05 ± 0.50	**99.95 ± 0.05**	99.66 ± 0.17	95.98 ± 0.38	98.96 ± 0.41	99.63 ± 0.10	99.59 ± 0.13
4	94.53 ± 0.40	78.19 ± 4.07	96.87 ± 1.47	99.49 ± 0.37	99.38 ± 0.41	99.71 ± 0.23	98.77 ± 0.23	**99.61 ± 0.22**	98.51 ± 0.40	99.60 ± 0.18
5	24.68 ± 4.24	90.23 ± 0.90	93.58 ± 1.79	96.51 ± 2.71	98.98 ± 0.56	98.89 ± 1.77	99.52 ± 0.08	98.27 ± 1.16	**99.88 ± 0.13**	98.98 ± 0.87
6	25.75 ± 1.55	95.67 ± 0.92	64.30 ± 10.63	94.01 ± 0.59	**99.02 ± 0.44**	94.63 ± 3.51	92.39 ± 0.25	95.36 ± 1.35	96.94 ± 0.99	98.04 ± 0.64
7	81.91 ± 0.98	86.79 ± 6.97	92.67 ± 3.54	97.77 ± 0.61	**99.79 ± 0.09**	98.79 ± 0.73	97.75 ± 0.27	98.01 ± 0.65	97.74 ± 0.70	99.77 ± 0.32
8	76.53 ± 0.84	97.78 ± 0.84	86.44 ± 6.39	98.36 ± 0.60	**99.34 ± 0.18**	99.13 ± 0.42	89.56 ± 0.80	99.12 ± 0.22	98.58 ± 0.56	99.34 ± 0.40
9	74.18 ± 1.40	37.50 ± 2.50	88.37 ± 1.92	98.17 ± 0.60	98.42 ± 0.29	98.47 ± 2.09	89.41 ± 0.21	98.78 ± 0.43	**99.41 ± 0.40**	98.75 ± 0.24
10	93.19 ± 0.39	90.40 ± 0.46	96.95 ± 0.83	98.65 ± 0.39	**99.57 ± 0.48**	99.48 ± 0.62	95.09 ± 0.10	99.37 ± 0.43	99.15 ± 0.65	99.53 ± 0.18
11	95.76 ± 0.34	94.68 ± 0.50	95.75 ± 1.81	98.83 ± 0.21	99.45 ± 0.20	99.50 ± 0.66	95.07 ± 0.29	**99.62 ± 0.10**	99.52 ± 0.39	99.54 ± 0.23
12	34.16 ± 2.91	80.02 ± 0.95	92.28 ± 5.97	98.54 ± 1.03	**99.99 ± 0.01**	99.46 ± 0.62	98.60 ± 0.12	96.53 ± 5.21	99.61 ± 0.46	99.49 ± 0.44
13	57.52 ± 1.42	**98.05 ± 0.44**	81.41 ± 7.14	92.39 ± 0.69	94.93 ± 1.05	95.27 ± 5.66	85.20 ± 1.08	94.23 ± 0.77	94.27 ± 0.96	97.04 ± 1.11
14	85.91 ± 0.36	97.48 ± 0.22	95.74 ± 1.36	98.89 ± 0.26	**99.84 ± 0.07**	99.43 ± 0.36	90.66 ± 0.08	99.11 ± 0.14	98.62 ± 0.47	99.53 ± 0.22
15	68.66 ± 0.96	91.71 ± 1.36	88.86 ± 4.86	89.01 ± 1.50	92.20 ± 2.17	98.21 ± 1.69	**96.49 ± 0.45**	94.00 ± 1.28	89.88 ± 4.81	91.99 ± 2.51
16	98.64 ± 0.02	89.89 ± 2.64	99.56 ± 0.41	99.76 ± 0.08	99.85 ± 0.06	99.95 ± 0.03	97.65 ± 0.00	99.88 ± 0.01	99.86 ± 0.02	**99.92 ± 0.05**
OA (%)	87.53 ± 0.10	91.13 ± 0.24	95.28 ± 1.10	98.81 ± 0.13	99.48 ± 0.03	99.28 ± 0.48	94.05 ± 0.13	99.14 ± 0.12	99.10 ± 0.13	**99.53 ± 0.15**
AA (%)	73.05 ± 0.28	86.53 ± 0.62	91.40 ± 2.06	97.37 ± 0.26	98.77 ± 0.12	98.72 ± 0.58	94.03 ± 0.15	98.10 ± 0.40	98.13 ± 0.42	**98.79 ± 0.29**
Kappa	85.37 ± 0.11	89.86 ± 0.28	94.48 ± 1.29	98.60 ± 0.15	99.40 ± 0.03	99.15 ± 0.56	93.03 ± 0.15	98.99 ± 0.13	98.95 ± 0.15	**99.45 ± 0.18**

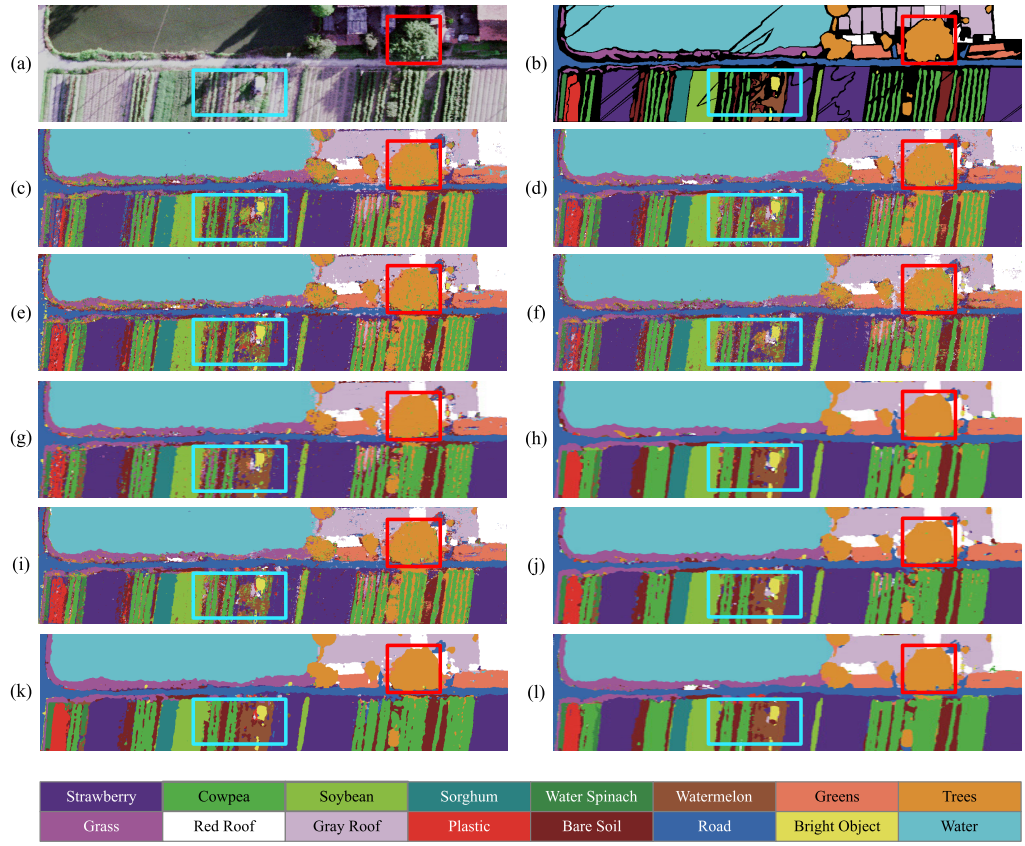
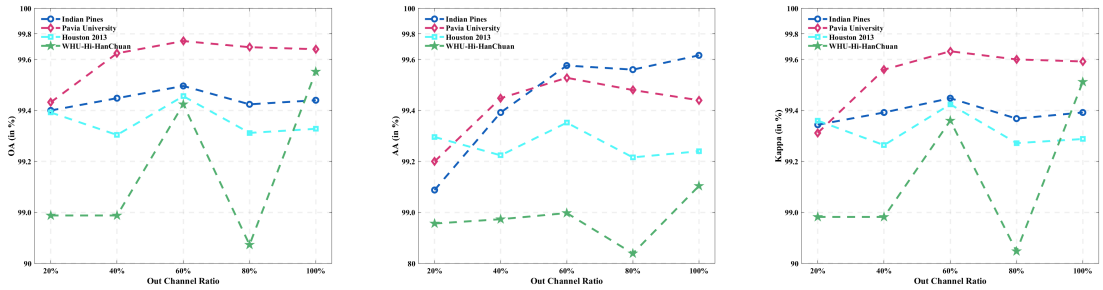


Fig. 9. Classification maps obtained by the compared methods on the WHU-Hi-HanChuan dataset. (a) False-color image. (b) Ground truth. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) SSFTT. (g) CASST. (h) DSFormer. (i) FDGC. (j) SGMAE. (k) WFCG. (l) DAGCN.

in terms of OA, AA, and the Kappa coefficient. Among the 16 land cover categories, the proposed method achieves higher classification accuracy than other methods in the water category. The SVM method only achieved an OA of 87.53%. Among the two CNN-based classification models, 3D-CNN still performed best with an OA of 95.28%. Among the three Transformer-based methods, CASST achieved the highest performance with an OA of 99.48%. Among the three GCN-based methods, SGMAE performed best with an OA of 99.14%. Our proposed DAGCN achieves the most optimal results, with an OA of 99.53%, an AA of 98.79%, and a $\mathcal{K}$ of 99.45.

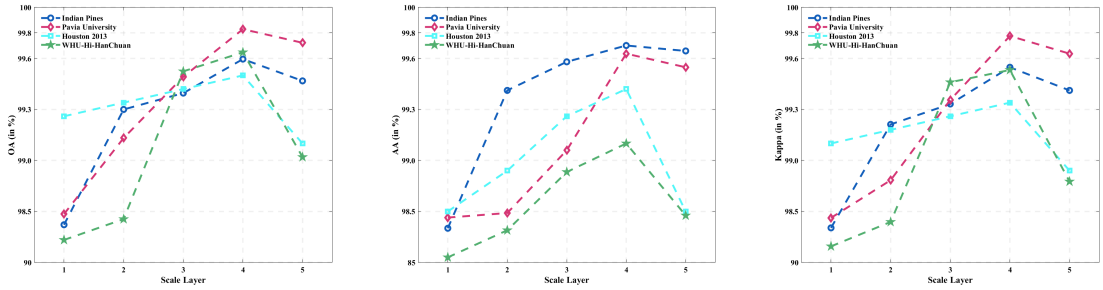
Fig. 9 clearly displays classification results from different methods for the dataset. For comparison, we present the corresponding WHU-Hi-HanChuan's pseudo-color images

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(a) (b) (c)

Fig. 10. Sensitivity analysis for the proposed method with different output dimensions in terms of (a) OA, (b) AA, and (c) Kappa.



(a) (b) (c)

Fig. 11. Sensitivity analysis for the proposed method with different multiscale layer counts in terms of (a) OA, (b) AA, and (c) Kappa.

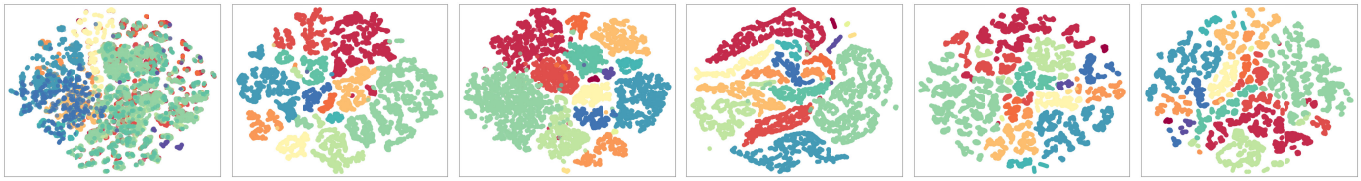
alongside its ground-truth labels. Evidently, SVM consistently exhibits the poorest classification performance overall. Significant accuracy gaps clearly exist among methods being compared in the red- and blue-boxed regions. Notably, SVM and both CNN-based approaches still perform poorly in both highlighted areas. Excluding FDGC, all other methods generally achieve good classification results in the red-boxed area. For the blue-boxed area, CASST relatively performs well among Transformer models. Among the three GCN-based methods, WFCG indeed achieves higher OA but performs poorly in this specific region. In contrast, the proposed DAGCN achieves the highest test accuracy, thus demonstrating good performance in both marked areas.

D. Discussion and Analysis

1) Sensitivity to Different Output Dimensions: Ablation experiments explored how RAGC feature output dimensions (0.2× to 1.0× band group dimension with 0.2× increments) affect classification performance, with results for OA, AA, and Kappa across four datasets shown in Fig. 10. Except Houston 2013 (a dip at 0.4×), all datasets show steady performance improvement from 0.2× to 0.6×, a decline at 0.8× and a rebound at 1.0×; Houston 2013, Pavia University have small variations with peaks at 0.6×, while WHU-Hi-HanChuan fluctuates more and peaks at 1.0×, due to its low interband correlation, high noise and uneven class distribution requiring more feature dimensions. 0.6× yields optimal/near-optimal performance for three datasets, revealing an intrinsic low-dimensional discriminative subspace in hyperspectral data's high-dimensional manifolds, with this dimension achieving high classification accuracy (e.g., 99.62% OA for Indian Pines) by acting as implicit regularization to balance expressiveness and overfitting, forcing compact, generalized representations and discarding redundant noisy information—critical for high-dimensional, redundant HSI data. Though 1.0× is optimal for WHU-Hi-HanChuan, 0.6× is near-optimal for it and optimal for others, so 0.6× band group dimension is adopted as the unified output dimension for all datasets in subsequent experiments.

2) Sensitivity to Different Multiscale Layer Counts: Ablation experiments on the MAGC module with layer counts $K \in \{1, 2, 3, 4, 5\}$ are shown in Fig. 11, it reveals consistent trends across all datasets: classification performance improves as K increases from 1 to 4, then declines at $K = 5$. For Indian Pines, OA increases from 96.29% ($K = 1$) to 99.62% ($K = 4$), then decreases to 99.46%; AA rises from 94.06% to 99.72% to 99.68%; Kappa improves from 95.77% to 99.56% to 99.39%. Pavia University shows OA peaking at 99.84% (from 98.07%), AA at 99.66% (from 96.89%), and Kappa at 99.79% (from 97.44%). Houston 2013's OA improves from 99.21% to 99.52%, then falls to 99.01%; AA increases from 98.50% to 99.42% to 98.51%; Kappa rises from 99.02% to 99.31% to 98.82%. WHU-Hi-HanChuan exhibits pronounced gains: OA jumps from 93.76% to 99.67% to 98.90%; AA surges from 86.36% to 99.00% to 97.44%; Kappa increases from 92.69% to 99.54% to 98.72%. Four layers provide optimal multiscale representation across all datasets; the performance decline at five layers likely results from over-smoothing effects where excessive contextual information overwhelms discriminative features, leading to the selection of $K = 4$ for the MAGC module in DAGCN.

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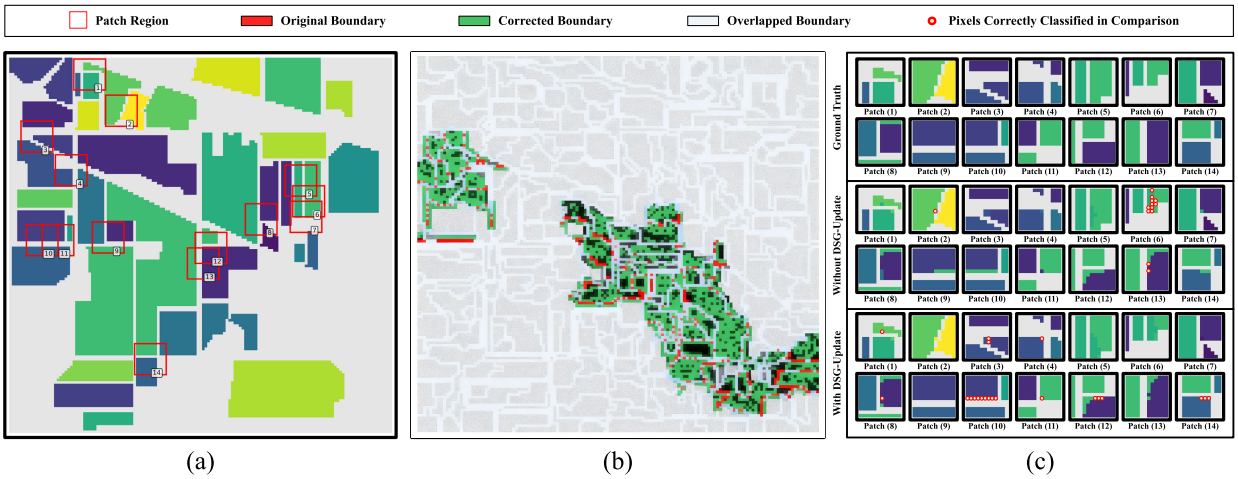


(a) (b) (c) (d) (e) (f)

Fig. 12. T-SNE visualizations of feature representations aggregated by different methods. (a) Original data. (b) FDGC. (c) SGMAE. (d) WFCG. (e) RAGC (our work, without multi-scale layer). (f) MAGC (our work, with multi-scale layer).

TABLE VI
EFFECTIVENESS ANALYSIS OF DSG UPDATE AND MSCA ON ALL DATASETS IN TERMS OF OA (%), AA (%), AND $\mathcal{K}$

DSG Update	MSCA	Indian Pines			Pavia University			Houston 2013			WHU-Hi-HanChuan		
		OA	AA	$\mathcal{K} \times 100$	OA	AA	$\mathcal{K} \times 100$	OA	AA	$\mathcal{K} \times 100$	OA	AA	$\mathcal{K} \times 100$
✓		99.57	99.70	99.51	99.78	99.55	99.71	99.53	99.41	99.49	99.21	98.12	99.07
	✓	99.50	99.67	99.43	99.80	99.50	99.74	99.51	99.39	99.47	97.27	94.10	96.78
✓	✓	99.62	99.72	99.56	99.84	99.66	99.79	99.57	99.44	99.53	99.53	98.79	99.45



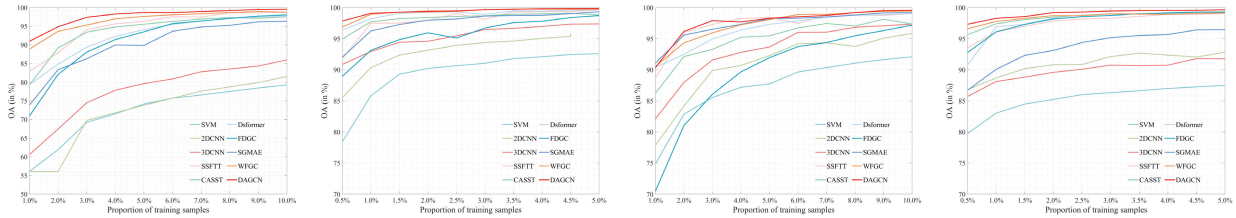
(a) (b) (c)

Fig. 13. Detailed illustration of ablation experiment results for the DSG Update module on the Indian Pines dataset. (a) Highlights patch regions demonstrating classification performance differences between configurations with and without the DSG Update module. (b) Changes in superpixel boundaries before and after ablation. Ground-truth classifications and (c) classification prediction images with and without DSG Update for each patch region.

3) Effectiveness of MAGC: This section evaluates the effectiveness of MAGC. The t-SNE plots of different methods on the Indian Pine dataset are shown in Fig. 12. It can be observed that the features extracted by FDGC, SGMAE, WFCG, and the independent RAGC exhibit low discriminative ability. Specifically, the features extracted by FDGC and RAGC lack intraclass compactness. Although the features extracted by SGMAE and WFCG exhibit intraclass compactness, their interclass discriminative ability is insufficient, making it difficult for the model to find clear decision boundaries to distinguish between different classes. In contrast, the features extracted by MAGC exhibit clear discriminative ability and perform well in terms of intraclass compactness and interclass separability. This shows that although SGMAE employs a self-supervised graph mask autoencoder framework to learn multiscale spatial–spectral features from unlabeled data through masking and reconstruction, extracting global context and local details with adaptive weight fusion, its ability to model complex nonlinear relationships between adjacent nodes remains limited in HSI classification tasks. The WFCG method innovatively combines the local feature extraction ability of CNNs with the global relationship modeling advantage of GATs. However, when faced with the rich spatial–spectral features of HSI, it may exhibit limitations due to its fixed graph structure or single attention mechanism. Although FDGC overcomes the limitations of traditional GCNs with static topology by performing dynamic feature aggregation through a trainable adjacency matrix, its parallel architecture still faces challenges in deeply integrating global and local information. RAGC introduces an adaptive aggregation mechanism to adaptively capture complex dependencies between nodes. At the same time, the addition of multi-scale layer based on RAGC makes MAGC more effective in learning more discriminative feature representations.

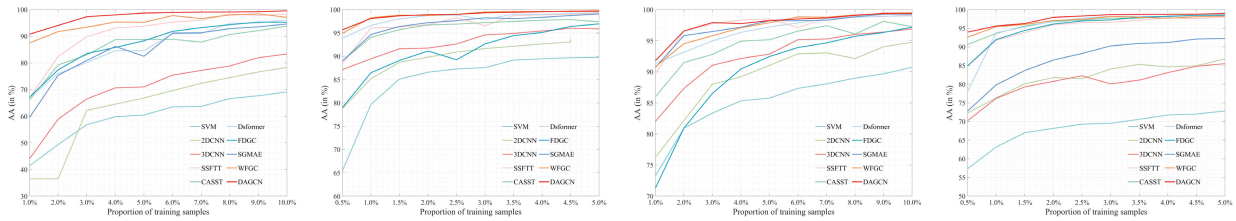
4) Effectiveness of STGC Strategy: This section systematically evaluates the effectiveness of the STGC strategy. The classification performance across four datasets is shown in Table VI. It can be seen that both DSG Update and MSCA

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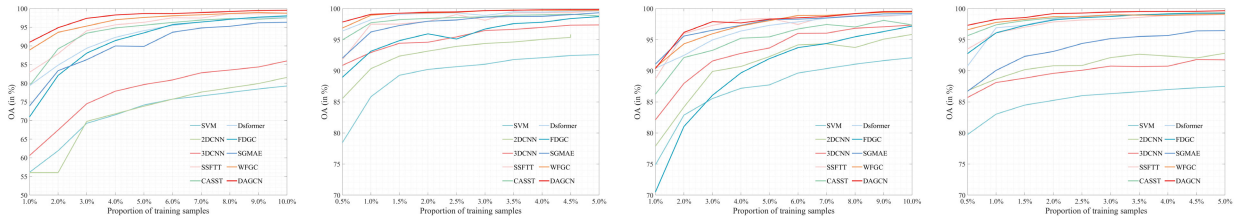
(a) (b) (c) (d)

Fig. 14. Sensitivity analysis was conducted on OA of all compared methods for four datasets: (a) Indian Pines, (b) Pavia University, (c) Houston 2013, and (d) WHU-Hi-HanChuan, at different training ratios.



(a) (b) (c) (d)

Fig. 15. Sensitivity analysis was conducted on AA of all compared methods for four datasets: (a) Indian Pines, (b) Pavia University, (c) Houston 2013, and (d) WHU-Hi-HanChuan, at different training ratios.



(a) (b) (c) (d)

Fig. 16. Sensitivity analysis was conducted on Kappa of all compared methods for four datasets: (a) Indian Pines, (b) Pavia University, (c) Houston 2013, and (d) WHU-Hi-HanChuan, at different training ratios.

independently achieve significant improvements in classification performance. Fig. 13 presents the ablation results of the DSG Update module on the Indian Pines dataset, demonstrating its effective correction of edge pixel classification errors caused by jagged superpixel segmentation boundaries. The module's error correction has practical bounds; it can recover moderate segmentation inaccuracies through temporal edge smoothing and re-assignment, but recovery is limited when initial segmentation completely fails to isolate a small target, as the necessary spatial–spectral context is lost. Furthermore, the integrated model STGC strategy further enhances the classification capabilities, exhibiting strong complementarity and robustness across all datasets. This improvement stems from the hybrid architecture's ability to simultaneously update multitemporal image structures while efficiently extracting discriminative features across phases, thereby capturing key discriminative information embedded in different spectral bands. Overall, both DSG Update and MSCA contribute to classification, and their integration yields the optimal results across diverse scenarios.

E. Sensitivity to Different Training Ratios

To validate the robustness of the proposed DAGCN framework, a comprehensive set of experiments was conducted by modulating the training sample proportions on four representative HSI datasets, including Indian Pines, Pavia University, Houston 2013, and WHU-Hi-HanChuan. The performance dynamics of all comparative methods under different training ratios are illustrated in Figs. 14–16, where the curve for the proposed DAGCN is highlighted in red for clear visualization. Specifically, the training sample ratios configured for the four datasets span the ranges of 1.0%–10.0% (with a step size of 1.0%), 0.5%–5.0% (with a step size of 0.5%), 1.0%–10.0% (with a step size of 1.0%), and 0.5%–5.0% (with a step size of 0.5%), respectively. Experimental findings demonstrate that the classification accuracy of the DAGCN framework exhibits a steady upward trend with the increase in training sample proportion. Furthermore, DAGCN outperforms all other state-of-the-art comparative methods at the majority of training ratios across the four datasets, which further corroborates the robust performance and superior effectiveness of the designed hybrid network architecture.

V. Conclusion

We propose a DAGCN framework for HSI classification, which effectively addresses the challenge of rigid graph structures. To mitigate spectral–spatial decoupling and noise sensitivity, we introduce a DGS strategy that maps

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high-dimensional spectral slices to a multigraph space, enabling a transition from pixel- to region-level prototypes. To address spatial adaptability and insufficient feature granularity, MAGC employs region-aware convolutions to learn neighborhood information. It integrates hybrid pooling to capture multiscale micro-variations, achieves scale alignment through upsampling, and preserves pixel details via residual connections. To mitigate spatiotemporal feature drift and boundary misalignment, the STGC strategy enhances multi-image topological continuity through morphological extraction and centroid redistribution. It further employs MSCA to select discriminative spectral band features. Extensive experiments demonstrate that DAGCN effectively optimizes graph structures and enhances classification performance, outperforming nine state-of-the-art methods across four HSI benchmark datasets. It achieves synergistic improvements in efficiency, accuracy, and robustness. Future research directions include further optimizing dynamic graph update strategies and developing more discriminative feature learning methods to achieve higher-precision HSI classification.

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